

Exact block reductions in Fourier-modal scattering theory for layered periodic media

N. V. Popov¹

¹Department of Physics, affiliation to be specified

Abstract

The Fourier modal method represents Maxwell's equations in a periodic layer by a finite set of n reciprocal-lattice harmonics. In a generic formulation, the modal problem has dimension $2n$ and the stable composition of layers by a Redheffer product requires dense cubic algebra. We derive exact reductions of these operators for structures occurring frequently in diffraction theory. For an isotropic homogeneous layer, a permutation to the per-harmonic basis $(E_{x,h}, E_{y,h})$ decomposes the Maxwell operator into independent 2×2 blocks at arbitrary incidence. Consequently, an arbitrary homogeneous multilayer is propagated in $O(nN_u)$ operations, where N_u is the number of homogeneous interfaces. For a y -invariant patterned layer, the modal matrix is block diagonal at normal incidence and block lower triangular under conical illumination. Its spectrum is therefore obtained from two $n \times n$ non-Hermitian eigenproblems; in the conical case we give an explicit reconstruction of the coupled eigenvectors. We also derive a block contraction for quadratic volume observables that avoids the $4n \times 4n$ intermediate matrices of the direct expression. Numerical experiments against grcwa and S4 verify the scattering observables and Fourier fields to the accuracy of the reference discretization. For an extreme- ultraviolet absorber on a 40-period Mo/Si mirror, the homogeneous-stack reduction accelerates the scattering stage by approximately 3.4–3.7, while the combined one-dimensional reductions give an order-of-magnitude or greater acceleration for line gratings. The analysis separates exact operator reductions from approximation-dependent convergence acceleration and identifies the patterned-layer eigensystem as the remaining asymptotic bottleneck.

Keywords: Fourier modal method; rigorous coupled-wave analysis; diffraction gratings; scattering matrix; conical incidence; block-triangular eigenproblem.

1 Introduction

Rigorous coupled-wave analysis (RCWA), also called the Fourier modal method (FMM), is the workhorse for computing the optical response of layered, laterally periodic photonic structures: diffraction gratings, photonic crystals, thin-film solar cells, metasurfaces, and the phase-shifting absorber stacks of extreme-ultraviolet (EUV) lithography [1–3]. The method expands the electromagnetic field in a truncated set of n planewave (Fourier) harmonics in the plane of periodicity, reduces Maxwell's equations in each laterally structured layer to a $2n \times 2n$ non-Hermitian eigenproblem, and propagates the modal amplitudes across the stack with a scattering-matrix (Redheffer) recursion [4, 5]. Reflectivity, transmissivity, per-order diffraction efficiencies, near-field intensities, and absorbed-power distributions all follow from the assembled S -matrix and the per-layer modal data.

The numerical difficulty is controlled by the number n of retained spatial harmonics. A patterned layer requires a non-Hermitian eigenproblem of order $2n$, while stable composition across interfaces requires dense matrix products and linear solves of the same order. Both stages scale cubically and store $O(n^2)$ complex numbers. Large-period gratings make this cost especially severe: at wavelength λ , a period L_x

admits on the order of $2L_x/\lambda$ propagating diffraction channels before evanescent harmonics are included. Repeated evaluations in wavelength, angle, or geometric parameter studies compound the cost.

A generic implementation spends its time in two algebraic operations:

1. $O((2n)^3)$ for the (generally non-Hermitian, complex) eigendecomposition (23) of each laterally structured layer, and
2. $O((2n)^3)$ dense matrix products and inversions *per layer interface* in the Redheffer recursion (37).

As a representative geometry, consider vacuum | absorber | cap | $40\times(\text{Mo/Si})$ | substrate. The single patterned layer is followed by approximately 80 homogeneous interfaces. Treating those interfaces as generic dense scatterers repeats cubic work even though translational invariance forbids coupling between distinct in-plane wavevectors in every homogeneous layer. The purpose of this paper is to expose that invariance, and related invariant subspaces of the patterned operator, directly in the finite-dimensional Maxwell system.

1.1 Context and main results

The Fourier-modal method was introduced for diffraction gratings by Moharam and Gaylord [1] and refined into a stable, efficient formulation for binary gratings in [2]. The stability of the layer-recursion step was settled by Li's comparison of the R -, T -, and S -matrix algorithms [5], establishing the scattering-matrix (Redheffer) formulation as the method of choice for thick and lossy stacks; our work uses exactly this recursion [6]. The convergence of the modal expansion for discontinuous media was analyzed by Li [7], whose inverse rule (and its generalization, the normal-vector field / fast Fourier factorization [8]) removes the algebraic convergence bottleneck of the naive Laurent rule for high-contrast metallic gratings. The scattering-matrix treatment of patterned multilayer photonic structures was systematized by Whittaker and Culshaw [4].

Convergence acceleration by Fourier factorization and normal-vector fields reduces the harmonic order needed at a specified accuracy [7, 8]. The present work addresses a different question: given a fixed Fourier discretization, which parts of its algebra are exactly reducible? The principal results are:

1. A homogeneous isotropic layer is a direct sum of n two-dimensional polarization spaces, for any reciprocal-lattice truncation and any Bloch wavevector. A homogeneous stack therefore admits an exact per-harmonic scattering recursion.
2. For a y -invariant isotropic profile, the patterned modal matrix splits into two n -dimensional spectral problems. At normal incidence the matrix is block diagonal; at conical incidence it is block lower triangular, and the off-diagonal eigenvector block is obtained from a Sylvester equation.
3. Quadratic volume observables can be contracted using $2n$ -dimensional blocks, without constructing the direct $4n$ -dimensional expression.

These are identities of the discretized Maxwell operator. They introduce no material, weak-coupling, or paraxial approximation and are complementary to factorization rules that improve convergence with n .

The open-source solvers `grcwa` [9] and `S4` [3] are used as independent numerical references. The derivation is developed in sections 2 to 4; implementation and numerical tests are reported in sections 8 and 9.

2 Fourier-modal formulation

The plan of this section, in words. The goal here is to turn Maxwell's equations in a layered periodic medium into a finite set of matrix operations that a computer can diagonalize and compose. Three ideas do

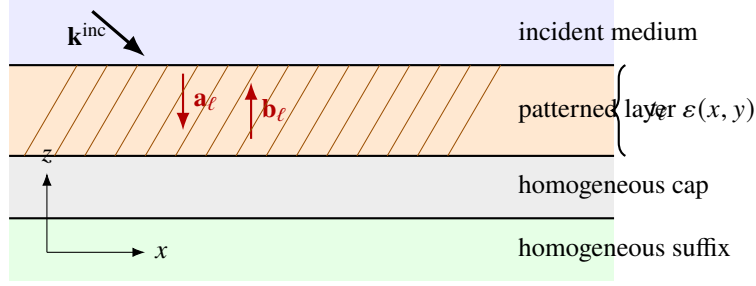


Figure 1: Layered periodic medium and modal-amplitude convention. Only the patterned layers couple distinct reciprocal-lattice harmonics. Every homogeneous region retains polarization coupling within a fixed harmonic but does not couple different harmonics.

the work, and each subsection adds one. *First*, periodicity in the plane lets every field be written as a sum of n plane waves (the “harmonics”), so the derivatives ∂_x, ∂_y become multiplication by known numbers and the only nontrivial direction left is z . *Second*, the z -components of the fields are not independent — they are fixed algebraically by the four transverse components — so Maxwell’s equations reduce to a first-order system in z for those four components, equivalently an eigenproblem whose eigenvalues are the squared vertical wavevectors q_j^2 of the layer’s modes. *Third*, the layers are joined by matching the transverse fields at each interface, which is expressed compactly as a scattering-matrix (Redheffer) recursion. The reductions of Sections 3 and 4 are then, at heart, just “notice the block structure in these matrices and stop multiplying zeros.”

2.1 Geometry, harmonics, and notation

Consider a stack of layers, each invariant along z (the propagation axis), with the in-plane permittivity $\varepsilon(x, y)$ periodic on a lattice with direct vectors $\mathbf{L}_1, \mathbf{L}_2$ and reciprocal vectors $\mathbf{K}_1, \mathbf{K}_2$. The field is expanded in the n retained harmonics $\{\mathbf{G}_i\}_{i=1}^n \subset \{m_1\mathbf{K}_1 + m_2\mathbf{K}_2\}$, so that every in-plane quantity is represented by n Fourier coefficients and the transverse wavevectors of harmonic i are

$$k_{x,i} = k_{x0} + (m_1\mathbf{K}_1 + m_2\mathbf{K}_2) \cdot \hat{x}, \quad k_{y,i} = k_{y0} + (m_1\mathbf{K}_1 + m_2\mathbf{K}_2) \cdot \hat{y}, \quad (1)$$

with k_{x0}, k_{y0} fixed by the incidence angles. We write $\mathbf{k}_x, \mathbf{k}_y \in \mathbb{C}^n$ for the diagonal entries $k_{x,i}, k_{y,i}$, and let $\mathbf{q} \in \mathbb{C}^{2n}$ denote the (unknown) vertical propagation constants. The total block size is $N = 2n$; all scattering matrices are $N \times N$.

Figure 1 fixes the geometry and the amplitude convention used throughout the paper. Within layer ℓ , $\mathbf{a}_\ell$ denotes modes whose phase advances in the positive z direction and $\mathbf{b}_\ell$ denotes modes whose phase advances in the negative z direction. For evanescent waves the branch convention introduced below turns these into decaying solutions in the corresponding half space.

For clarity, two orderings of the $2n$ transverse components will be used. The *component-major* ordering is

$$\mathbf{u}_c = [u_{x,1}, \dots, u_{x,n}, u_{y,1}, \dots, u_{y,n}]^\top, \quad (2)$$

whereas the *harmonic-major* ordering is

$$\mathbf{u}_h = [u_{x,1}, u_{y,1}, u_{x,2}, u_{y,2}, \dots, u_{x,n}, u_{y,n}]^\top. \quad (3)$$

They are related by a permutation matrix Π , with $\mathbf{u}_h = \Pi\mathbf{u}_c$ and $\Pi^{-1} = \Pi^\top$. No approximation is associated with this change of basis; it merely makes different sparsity patterns visible.

2.2 The layer eigenproblem

We derive the eigenproblem from Maxwell's equations. With the $e^{+i\omega t}$ time convention, the curl equations in a source-free, non-magnetic medium are

$$\nabla \times \mathbf{E} = -i\omega \mathbf{H}, \quad \nabla \times \mathbf{H} = i\omega \varepsilon \mathbf{E}. \quad (4)$$

For a Bloch wave with in-plane Bloch vector $\mathbf{k}_{\parallel}^{(0)}$, a field component f has the truncated expansion

$$f(x, y, z) = \sum_{j=1}^n f_j(z) \exp \left[i(\mathbf{k}_{\parallel}^{(0)} + 2\pi \mathbf{G}_j) \cdot \mathbf{r}_{\parallel} \right]. \quad (5)$$

Orthogonality of these plane waves turns ∂_x and ∂_y into multiplication by iK_x and iK_y . Multiplication by a periodic permittivity is not diagonal: if $\varepsilon(\mathbf{r}_{\parallel}) = \sum_{\mathbf{g}} \hat{\varepsilon}_{\mathbf{g}} e^{2\pi i \mathbf{g} \cdot \mathbf{r}_{\parallel}}$, then

$$(\varepsilon f)_i = \sum_{j=1}^n \hat{\varepsilon}_{\mathbf{G}_i - \mathbf{G}_j} f_j, \quad C_{ij} = \hat{\varepsilon}_{\mathbf{G}_i - \mathbf{G}_j}. \quad (6)$$

This convolution is the sole source of coupling between distinct harmonics in a layer.

Fourier-expanding the transverse field components $e_x, e_y, h_x, h_y \in \mathbb{C}^n$ and stacking $\mathbf{e}_t = [e_x; e_y]$, $\mathbf{h}_t = [h_x; h_y]$ (in the ordering $[E_{x,1} \dots E_{x,n}; E_{y,1} \dots E_{y,n}]$), the in-plane derivatives ∂_x, ∂_y become multiplications by the diagonal matrices $K_x = \text{diag}(k_{x,i})$, $K_y = \text{diag}(k_{y,i})$, and the permittivity becomes the convolution operator C .

2.2.1 The curl equations in the Fourier basis

We derive the eigenproblem in full, so that every sign and prefactor is traceable to a scalar Maxwell equation. Writing the two curl equations (4) component by component and replacing $\partial_x \rightarrow iK_x$, $\partial_y \rightarrow iK_y$, $\partial_z \rightarrow d/dz$, the six scalar equations are

$$iK_y e_z - \frac{de_y}{dz} = -i\omega h_x, \quad (7)$$

$$\frac{de_x}{dz} - iK_x e_z = -i\omega h_y, \quad (8)$$

$$iK_x e_y - iK_y e_x = -i\omega h_z, \quad (9)$$

$$iK_y h_z - \frac{dh_y}{dz} = i\omega C e_x, \quad (10)$$

$$\frac{dh_x}{dz} - iK_x h_z = i\omega C e_y, \quad (11)$$

$$iK_x h_y - iK_y h_x = i\omega C e_z. \quad (12)$$

Here and below K_x, K_y are the $n \times n$ diagonal matrices $K_x = \text{diag}(k_{x,i})$, $K_y = \text{diag}(k_{y,i})$, and C is the $n \times n$ permittivity convolution matrix (6). The only coupling between different harmonics enters through C ; every other operator in this layer problem is diagonal.

Longitudinal components. The z -components of the curl equations involve neither z -derivatives nor the longitudinal field pairs, so they determine e_z and h_z algebraically. From (12) one reads $K_x h_y - K_y h_x = \omega C e_z$, i.e.

$$e_z = -\frac{1}{\omega} C^{-1} (K_y h_x - K_x h_y), \quad (13)$$

and from (9),

$$h_z = \frac{1}{\omega} (K_y e_x - K_x e_y). \quad (14)$$

The inverse C^{-1} is the inverse-permittivity operator; its matrix representation $\mathbf{E}$ is defined in (25) below and is the only place where a matrix inverse enters the layer construction.

The four tangential equations. Substituting (13)–(14) into the tangential curl equations (7)–(11) and solving for the z -derivatives eliminates the longitudinal components completely. Take them one at a time.

Equation (8). Using $e_z = -\frac{1}{\omega}C^{-1}(K_y h_x - K_x h_y)$,

$$\frac{de_x}{dz} = -i\omega h_y + iK_x e_z = -i\omega h_y - \frac{i}{\omega}K_x C^{-1}K_y h_x + \frac{i}{\omega}K_x C^{-1}K_x h_y. \quad (15)$$

Equation (7).

$$\frac{de_y}{dz} = iK_y e_z + i\omega h_x = i\omega h_x - \frac{i}{\omega}K_y C^{-1}K_y h_x + \frac{i}{\omega}K_y C^{-1}K_x h_y. \quad (16)$$

Equation (11). Using $h_z = \frac{1}{\omega}(K_y e_x - K_x e_y)$,

$$\frac{dh_x}{dz} = i\omega C e_y + iK_x h_z = \frac{i}{\omega}K_x K_y e_x + (i\omega C - \frac{i}{\omega}K_x^2)e_y. \quad (17)$$

Equation (10).

$$\frac{dh_y}{dz} = iK_y h_z - i\omega C e_x = (-i\omega C + \frac{i}{\omega}K_y^2)e_x - \frac{i}{\omega}K_y K_x e_y. \quad (18)$$

Collecting (15)–(18) gives the first-order system

$$\frac{d}{dz} \begin{bmatrix} e_x \\ e_y \\ h_x \\ h_y \end{bmatrix} = \mathbf{A} \begin{bmatrix} e_x \\ e_y \\ h_x \\ h_y \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{0} \end{bmatrix}, \quad (19)$$

with the $2n \times 2n$ blocks, read directly off (15)–(18),

$$\mathbf{A}_{12} = \begin{bmatrix} -\frac{i}{\omega}K_x C^{-1}K_y & -i\omega I + \frac{i}{\omega}K_x C^{-1}K_x \\ i\omega I - \frac{i}{\omega}K_y C^{-1}K_y & \frac{i}{\omega}K_y C^{-1}K_x \end{bmatrix}, \quad (20)$$

$$\mathbf{A}_{21} = \begin{bmatrix} \frac{i}{\omega}K_x K_y & i\omega C - \frac{i}{\omega}K_x^2 \\ -i\omega C + \frac{i}{\omega}K_y^2 & -\frac{i}{\omega}K_y K_x \end{bmatrix}. \quad (21)$$

The off-diagonal placement is deliberate: $d\mathbf{e}_t/dz = \mathbf{A}_{12}\mathbf{h}_t$ and $d\mathbf{h}_t/dz = \mathbf{A}_{21}\mathbf{e}_t$. Each block in (21) is the composition of a diagonal kinematic factor (K_x, K_y, ω) with either C or C^{-1} ; only these two non-diagonal factors couple harmonics, and they do so *only* in $\mathbf{A}_{12}$ (through C^{-1}) and in the diagonal terms of $\mathbf{A}_{21}$ (through C). This separation is the seed of all the block reductions in sections 3 and 4.

Modal spectrum. Because $\mathbf{A}$ is block-anticommuting, $\mathbf{A}^2 = \text{blkdiag}(\mathbf{A}_{12}\mathbf{A}_{21}, \mathbf{A}_{21}\mathbf{A}_{12})$, the eigenvalues of $\mathbf{A}$ are the $\pm$ -symmetric pairs $\{\pm\lambda_j\}$: if $[\mathbf{u}; \mathbf{w}]$ is an eigenvector with eigenvalue λ , then $\mathbf{A}_{12}\mathbf{A}_{21}\mathbf{u} = \lambda^2\mathbf{u}$ and $\mathbf{A}_{21}\mathbf{A}_{12}\mathbf{w} = \lambda^2\mathbf{w}$, and the two blocks $\mathbf{A}_{12}\mathbf{A}_{21}$ and $\mathbf{A}_{21}\mathbf{A}_{12}$ share their spectrum (the products AB and BA of two square matrices always have the same characteristic polynomial). Eliminating the magnetic component from (19) gives the second-order equation for the transverse electric field,

$$\frac{d^2\mathbf{e}_t}{dz^2} = \mathbf{A}_{12} \frac{d\mathbf{h}_t}{dz} = \mathbf{A}_{12}\mathbf{A}_{21} \mathbf{e}_t, \quad (22)$$

whose eigenvalues are $-q_j^2$ with $q_j \equiv \lambda_j/i$: the modal fields vary as $e^{\pm iq_j z}$, and the vertical propagation constants q_j are the square roots (with the branch cut of section 2) of the eigenvalues of (22). Writing $\lambda = iq_j$ recovers the pairing $\{\pm iq_j\}$ quoted in the introduction. The matrix $\mathbf{M}$ defined in (24) below is a $2n \times 2n$ representation of the same modal problem: its eigenvalues are the q_j^2 (each with multiplicity two), i.e. $\mathbf{M}$ is similar to $-\mathbf{A}_{12}\mathbf{A}_{21}$. We use the $\mathbf{M}$ (amplitude) representation throughout, because it renders

the interface matrices (29) and the field reconstruction diagonal in the per-harmonic basis (see section 3); the spectra of the two representations coincide exactly. In the planewave (Fourier-modal) basis the same problem is written as the $2n \times 2n$ non-Hermitian eigenproblem

$$\mathbf{M} \Phi = \Phi \text{diag}(\mathbf{q}^2), \quad (23)$$

whose eigenvalues are the squared vertical propagation constants q_j^2 , and the modal fields vary as $e^{\pm i q_j z}$. The matrix $\mathbf{M}$ is built from the permittivity convolution (Toeplitz) matrix $C_{ij} = \hat{\varepsilon}(\mathbf{G}_i - \mathbf{G}_j)$,

$$\mathbf{M} = \text{ep2} \mathbf{K} - \mathbf{k} \mathbf{k}^\top, \quad \text{ep2} = \text{blkdiag}(C, C), \quad (24)$$

with $\mathbf{E} = C^{-1}$, and

$$\mathbf{K} = \omega^2 \mathbf{I} - (\mathbf{J}_k \mathbf{E}) \mathbf{J}_k^\top, \quad \mathbf{J}_k = \begin{bmatrix} -\text{diag}(\mathbf{k}_y) \\ \text{diag}(\mathbf{k}_x) \end{bmatrix}, \quad \mathbf{k} = \begin{bmatrix} \text{diag}(\mathbf{k}_x) \\ \text{diag}(\mathbf{k}_y) \end{bmatrix}. \quad (25)$$

The branch cut $\mathbf{q}_i \mapsto -\mathbf{q}_i$ when $\text{Im } \mathbf{q}_i < 0$ selects the physically decaying (evanescent) modes in $+z$. Here ω is the (possibly complex) angular frequency; the lossy case $\text{Im } \omega \neq 0$ or complex ε is included throughout. For a homogeneous layer of permittivity ε_0 , $\mathbf{E} = 1/\varepsilon_0$ and the same formulas apply with C replaced by the scalar ε_0 . (The placement of $\mathbf{E}$ in (25) is consistent with the inverse-factorization treatment of the discontinuous normal field component [7]; alternative Fourier-factorization rules may be substituted without changing the block arguments below.)

2.3 S-matrix (Redheffer) recursion

2.3.1 Field representation and interface matrices

To connect two adjacent layers we need the modal fields expressed in terms of the amplitudes. The transverse fields at a height z inside layer ℓ are

$$\mathbf{E}_t = \mathbf{K}_\ell \Phi_\ell \frac{\mathbf{a} - \mathbf{b}}{\omega \mathbf{q}_\ell}, \quad \mathbf{H}_t = \Phi_\ell (\mathbf{a} + \mathbf{b}), \quad (26)$$

where $\mathbf{a}, \mathbf{b}$ are the forward/backward amplitude vectors propagated to the evaluation height (the forward amplitude is referenced at the left edge of the layer and the backward one at the right edge, so that within the layer $\mathbf{a}(z) = \mathbf{a} e^{i \mathbf{q}_\ell z}$, $\mathbf{b}(z) = \mathbf{b} e^{i \mathbf{q}_\ell (t_\ell - z)}$). Equation (26) is the reconstruction used in section 8; it is the modal-amplitude normalization in which the interface matrices below take their diagonal form. The transverse pair $(\mathbf{E}_t, \mathbf{H}_t)$ here is a fixed transverse rotation $(E_x, E_y) \leftrightarrow (E_y, -E_x)$ of the physical field (the rotation that makes the z -Poynting flux a plain scalar product); being identical on both sides of every interface, the rotation cancels from the continuity conditions (27), so the interface matrices below are those of the physical fields.

At an interface between layers ℓ and $\ell + 1$ the tangential components $\mathbf{E}_t, \mathbf{H}_t$ are continuous. Writing $\mathbf{u} = \mathbf{a} + \mathbf{b}$ (the $\mathbf{H}$ -combination) and $\mathbf{v} = \mathbf{a} - \mathbf{b}$ (the $\mathbf{E}$ -combination), continuity gives

$$\Phi_\ell \mathbf{u}_\ell = \Phi_{\ell+1} \mathbf{u}_{\ell+1}, \quad \mathbf{K}_\ell \Phi_\ell \frac{\mathbf{v}_\ell}{\mathbf{q}_\ell} = \mathbf{K}_{\ell+1} \Phi_{\ell+1} \frac{\mathbf{v}_{\ell+1}}{\mathbf{q}_{\ell+1}}. \quad (27)$$

Solving for the layer- $\ell + 1$ combinations,

$$\mathbf{u}_{\ell+1} = \Phi_{\ell+1}^{-1} \Phi_\ell \mathbf{u}_\ell = \mathbf{Q}_\ell^{-1} \mathbf{u}_\ell, \quad \mathbf{v}_{\ell+1} = \mathbf{q}_{\ell+1} (\mathbf{K}_{\ell+1} \Phi_{\ell+1})^{-1} \mathbf{K}_\ell \Phi_\ell \frac{1}{\mathbf{q}_\ell} \mathbf{v}_\ell = \mathbf{P}_\ell^{-1} \mathbf{v}_\ell, \quad (28)$$

with the interface matrices

$$\mathbf{Q}_\ell = \Phi_\ell^{-1} \Phi_{\ell+1}, \quad \mathbf{P}_\ell = \text{diag}(\mathbf{q}_\ell) (\mathbf{K}_\ell \Phi_\ell)^{-1} \mathbf{K}_{\ell+1} \Phi_{\ell+1} \text{diag}(1/\mathbf{q}_{\ell+1}), \quad (29)$$

and the transmission matrices $\mathbf{T}_{11} = \frac{1}{2}(\mathbf{Q} + \mathbf{P})$, $\mathbf{T}_{12} = \frac{1}{2}(\mathbf{Q} - \mathbf{P})$. In the (a, b) basis the map (28) reads

$$\begin{bmatrix} \mathbf{a}_{\ell+1} \\ \mathbf{b}_{\ell+1} \end{bmatrix} = \begin{bmatrix} \frac{1}{2}(\mathbf{Q}^{-1} + \mathbf{P}^{-1}) & \frac{1}{2}(\mathbf{Q}^{-1} - \mathbf{P}^{-1}) \\ \frac{1}{2}(\mathbf{Q}^{-1} - \mathbf{P}^{-1}) & \frac{1}{2}(\mathbf{Q}^{-1} + \mathbf{P}^{-1}) \end{bmatrix} \begin{bmatrix} \mathbf{a}_\ell \\ \mathbf{b}_\ell \end{bmatrix}, \quad (30)$$

and the reverse map, which we use below, is the inverse block matrix

$$\begin{bmatrix} \mathbf{a}_\ell \\ \mathbf{b}_\ell \end{bmatrix} = \begin{bmatrix} \mathbf{T}_{11} & \mathbf{T}_{12} \\ \mathbf{T}_{12} & \mathbf{T}_{11} \end{bmatrix} \begin{bmatrix} \mathbf{a}_{\ell+1} \\ \mathbf{b}_{\ell+1} \end{bmatrix}. \quad (31)$$

(That (31) is the inverse of (30) follows directly from $(\mathbf{Q} + \mathbf{P})(\mathbf{Q}^{-1} + \mathbf{P}^{-1}) + (\mathbf{Q} - \mathbf{P})(\mathbf{Q}^{-1} - \mathbf{P}^{-1}) = 4\mathbf{I}$.) We emphasize that (29)–(31) are *exact* identities obtained purely from the tangential field continuity; no small-parameter or paraxial approximation has been introduced.

2.3.2 The star product

The scattering matrix of the composite stack follows from the Redheffer star product [5, 6]. We derive the one-layer update in full. After layers $0, \dots, \ell$ have been processed the accumulated $S(0, \ell)$ satisfies, by (38) below,

$$\mathbf{a}_\ell = S_{11}\mathbf{a}_0 + S_{12}\mathbf{b}_\ell, \quad \mathbf{b}_0 = S_{21}\mathbf{a}_0 + S_{22}\mathbf{b}_\ell, \quad (32)$$

with $\mathbf{a}_\ell$ at the left edge of layer ℓ and $\mathbf{b}_\ell$ at its right edge. To incorporate layer $\ell + 1$: propagate $\mathbf{a}_\ell$ across layer ℓ (phase $\mathbf{d}_1 = \text{diag}(e^{iq_\ell t_\ell})$, so the forward amplitude at the interface is $\mathbf{d}_1\mathbf{a}_\ell$), cross the interface by the reverse map (31), and propagate the new backward amplitude back across layer $\ell + 1$ (phase $\mathbf{d}_2 = \text{diag}(e^{iq_{\ell+1}t_{\ell+1}})$, so the backward amplitude at the left edge of layer $\ell + 1$ is $\mathbf{d}_2\mathbf{b}_{\ell+1}$):

$$\mathbf{d}_1\mathbf{a}_\ell = \mathbf{T}_{11}\mathbf{a}_{\ell+1} + \mathbf{T}_{12}\mathbf{d}_2\mathbf{b}_{\ell+1}, \quad (33)$$

$$\mathbf{b}_\ell = \mathbf{T}_{12}\mathbf{a}_{\ell+1} + \mathbf{T}_{11}\mathbf{d}_2\mathbf{b}_{\ell+1}. \quad (34)$$

Eliminating $\mathbf{a}_\ell, \mathbf{b}_\ell$ between (32) and (34) is a linear solve for the new output amplitude $\mathbf{a}_{\ell+1}$. From the first relation in (34),

$$\mathbf{a}_\ell = \mathbf{d}_1^{-1}\mathbf{T}_{11}\mathbf{a}_{\ell+1} + \mathbf{d}_1^{-1}\mathbf{T}_{12}\mathbf{d}_2\mathbf{b}_{\ell+1}, \quad (35)$$

and substituting into (32) with $\mathbf{b}_\ell = \mathbf{T}_{12}\mathbf{a}_{\ell+1} + \mathbf{T}_{11}\mathbf{d}_2\mathbf{b}_{\ell+1}$,

$$(\mathbf{T}_{11} - \mathbf{d}_1 S_{12} \mathbf{T}_{12})\mathbf{a}_{\ell+1} = \mathbf{d}_1 S_{11}\mathbf{a}_0 + (\mathbf{d}_1 S_{12} \mathbf{T}_{11} - \mathbf{T}_{12})\mathbf{d}_2\mathbf{b}_{\ell+1}. \quad (36)$$

Solving (36) for $\mathbf{a}_{\ell+1}$ and substituting the result into (32) for $\mathbf{b}_0$ yields the update

$$\begin{aligned} \mathbf{P}_1 &= (\mathbf{T}_{11} - \mathbf{d}_1 S_{12} \mathbf{T}_{12})^{-1}, \\ \mathbf{S}_{11} &\leftarrow \mathbf{P}_1 \mathbf{d}_1 S_{11}, \quad \mathbf{S}_{12} \leftarrow \mathbf{P}_1 (\mathbf{d}_1 S_{12} \mathbf{T}_{11} - \mathbf{T}_{12}) \mathbf{d}_2, \\ \mathbf{S}_{21} &\leftarrow \mathbf{S}_{21} + \mathbf{S}_{22} \mathbf{T}_{12} \mathbf{S}_{11}, \quad \mathbf{S}_{22} \leftarrow \mathbf{S}_{22} (\mathbf{T}_{11} \mathbf{d}_2 + \mathbf{T}_{12} \mathbf{S}_{12}). \end{aligned} \quad (37)$$

Each update costs a handful of dense $N \times N$ matrix products plus one *LU*-based solve (the matrix inverse in $\mathbf{P}_1$ is evaluated as one factorization and two triangular solves).

The reason for using a scattering matrix rather than a transfer matrix is stability. A transfer matrix carries both exponentially growing and decaying solutions through a thick layer. Even when the physical answer is bounded, the intermediate transfer matrix can overflow or lose all significant digits. The scattering matrix instead relates incoming amplitudes to outgoing amplitudes and keeps the two directions separated. In the convention used in (37),

$$\begin{bmatrix} \mathbf{a}_{\ell+1} \\ \mathbf{b}_\ell \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}_\ell \begin{bmatrix} \mathbf{a}_\ell \\ \mathbf{b}_{\ell+1} \end{bmatrix}. \quad (38)$$

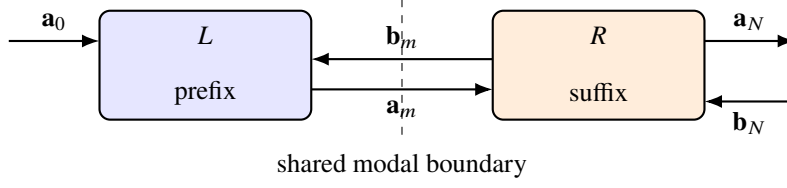


Figure 2: Redheffer composition as elimination of the two amplitudes at the shared internal boundary. The same diagram applies when the right subnetwork is represented compactly by independent 2×2 blocks.

Here $\mathbf{a}_{\ell+1}$ is the forward amplitude at the left edge of layer $\ell + 1$ (equivalently the right edge of layer ℓ) and $\mathbf{b}_{\ell+1}$ the backward amplitude at the right edge of layer $\ell + 1$, consistently with (26).

The update (37) is the specialization of a more general construction. If the stack is split into two subnetworks L and R that *share* a boundary (the “overlapping” case of Figure 2), introduce the internal amplitudes $\mathbf{a}_m, \mathbf{b}_m$ at the shared interface:

$$\mathbf{a}_m = L_{11}\mathbf{a}_0 + L_{12}\mathbf{b}_m, \quad (39)$$

$$\mathbf{b}_m = R_{21}\mathbf{a}_m + R_{22}\mathbf{b}_N. \quad (40)$$

Elimination gives

$$(I - L_{12}R_{21})\mathbf{a}_m = L_{11}\mathbf{a}_0 + L_{12}R_{22}\mathbf{b}_N. \quad (41)$$

Solving (41) for $\mathbf{a}_m$ and substituting into the outgoing relations gives the two-block cascade (49) of section 3.2. The Redheffer product is therefore not a heuristic matrix operation: in both forms (37) and (49) it is Gaussian elimination of the amplitudes at an internal interface.

3 Structure of homogeneous layers

3.1 A per-harmonic block lemma

In words, the lemma below says that a homogeneous layer is *diagonal in the harmonic index*: after re-ordering the components so that the two transverse components of each harmonic sit next to each other, the layer’s operator splits into n independent 2×2 pieces, one per diffraction order, and its modal basis is trivial (Φ is the identity). This single fact is what the per-harmonic suffix algorithm of Section 3.2 is built on.

Lemma 1. *Let the $2n$ field components be ordered by harmonic as $[E_{x,1}, E_{y,1}, E_{x,2}, E_{y,2}, \dots, E_{x,n}, E_{y,n}]$ (the per-harmonic interleaving). Then for a homogeneous isotropic layer of permittivity ε_0 ,*

$$\mathbf{K} = \text{blkdiag}(\mathbf{K}^{(1)}, \dots, \mathbf{K}^{(n)}), \quad \mathbf{K}^{(h)} = \omega^2 \mathbf{I}_2 - \frac{1}{\varepsilon_0} \begin{bmatrix} k_{y,h}^2 & -k_{x,h}k_{y,h} \\ -k_{x,h}k_{y,h} & k_{x,h}^2 \end{bmatrix}, \quad (42)$$

the propagation constants satisfy $\mathbf{q}_h = \mathbf{q}_{n+h} = q_h$ with $q_h^2 = \varepsilon_0 \omega^2 - k_{x,h}^2 - k_{y,h}^2$, and $\Phi = \mathbf{I}_{2n}$.

Proof. We first work in the component-major ordering, where the proof is a direct computation. From (25) with $E = 1/\varepsilon_0$ and $\mathbf{J}_k = [-\text{diag}(\mathbf{k}_y); \text{diag}(\mathbf{k}_x)]$, the product

$$\mathbf{J}_k \mathbf{J}_k^\top = \begin{bmatrix} -\text{diag}(\mathbf{k}_y) \\ \text{diag}(\mathbf{k}_x) \end{bmatrix} \begin{bmatrix} -\text{diag}(\mathbf{k}_y) & \text{diag}(\mathbf{k}_x) \end{bmatrix} = \begin{bmatrix} \text{diag}(\mathbf{k}_y^2) & -\text{diag}(\mathbf{k}_x \mathbf{k}_y) \\ -\text{diag}(\mathbf{k}_x \mathbf{k}_y) & \text{diag}(\mathbf{k}_x^2) \end{bmatrix} \quad (43)$$

is a 2×2 block matrix whose blocks are themselves *diagonal*. In the standard $[E_{x,1} \dots E_{x,n}; E_{y,1} \dots E_{y,n}]$ ordering, each harmonic h therefore couples only to itself: the (h, h) entry of the $(1, 1)$ -block, the $(h, n+h)$

entry of the $(1, 2)$ -block, etc. Forming the 2×2 submatrix of rows/columns $\{h, n+h\}$ for each h and applying the interleaving permutation that orders the components as $[E_{x,1}, E_{y,1}, E_{x,2}, E_{y,2}, \dots]$ yields exactly (42); different harmonics are never coupled.

It remains to verify the eigensystem. Because (42) is block diagonal, each harmonic obeys an independent dispersion relation. Writing the modal fields as $e^{i(q_h z)} e^{i(k_{x,h}x + k_{y,h}y)}$ and inserting into Maxwell's equations (4) with $\varepsilon = \varepsilon_0$ gives $k_{x,h}^2 + k_{y,h}^2 + q_h^2 = \varepsilon_0 \omega^2$, i.e. $q_h^2 = \varepsilon_0 \omega^2 - k_{x,h}^2 - k_{y,h}^2$. The corresponding eigenvectors are the coordinate vectors, so in the interleaved ordering $\Phi = \mathbf{I}_{2n}$ and the propagation constants $\mathbf{q} = [\mathbf{q}_x; \mathbf{q}_y]$ satisfy $\mathbf{q}_x = \mathbf{q}_y = q_h$ componentwise. $\square$

The permutation argument is central to the reduction. In the component-major basis the four $n \times n$ blocks are diagonal, but the two entries belonging to one diffraction order are separated in memory. Let Π denote the permutation from component-major to harmonic-major ordering. Then

$$\Pi \mathbf{K} \Pi^T = \text{blkdiag}(\mathbf{K}^{(1)}, \dots, \mathbf{K}^{(n)}). \quad (44)$$

Equation (44) is an identity, not an approximation. It shows directly why a homogeneous layer can be applied or inverted by n independent 2×2 operations. A dense implementation hides this fact by using the component-major ordering and consequently performs arithmetic on known zero entries.

The lemma is exact for *any* harmonic set — rectangular, circular, or anisotropic (n_x, n_y) — and for *any* incidence, including oblique ($k_{y0} \neq 0$): the only inputs are the per-harmonic pairs $(k_{x,h}, k_{y,h})$. It is the reason the quasi-1D “diagonal core” tricks known in the literature for $k_y = 0$ generalize directly to full 2D.

Corollary 2. *The interface matrices (29) between two homogeneous layers, and therefore the scattering matrix of any all-homogeneous stack, are 2×2 -block-diagonal in the interleaved ordering.*

Proof. For two homogeneous layers $\Phi_\ell = \Phi_{\ell+1} = \mathbf{I}_{2n}$, so the first interface matrix in (29) is $\mathbf{Q}_\ell = \mathbf{I}_{2n}$. The second is $\mathbf{P}_\ell = \text{diag}(\mathbf{q}_\ell) \mathbf{K}_\ell^{-1} \mathbf{K}_{\ell+1} \text{diag}(1/\mathbf{q}_{\ell+1})$. By Lemma 1 each of the four factors is block-diagonal with 2×2 blocks (the diag factors are diagonal, hence block-diagonal), and products and inverses of block-diagonal matrices are block-diagonal. Per harmonic h ,

$$\mathbf{P}^{(h)} = \frac{q_{\ell,h}}{q_{\ell+1,h}} (\mathbf{K}_\ell^{(h)})^{-1} \mathbf{K}_{\ell+1}^{(h)}, \quad (45)$$

with $\mathbf{K}^{(h)}$ as in (42). The transmission matrices $\mathbf{T}_{11} = \frac{1}{2}(\mathbf{I} + \mathbf{P})$, $\mathbf{T}_{12} = \frac{1}{2}(\mathbf{I} - \mathbf{P})$, the diagonal phase factors $\mathbf{d}_\ell$, and every matrix in the star update (37) are then block-diagonal as well: the inverse $(\mathbf{T}_{11} - \mathbf{d}_1 \mathbf{S}_{12} \mathbf{T}_{12})^{-1}$ of a block-diagonal matrix is block-diagonal, and all products preserve the structure. By induction over the recursion (37), the S -matrix of any all-homogeneous range is 2×2 -block-diagonal. $\square$

Each harmonic's 2×2 recursion is thus fully independent of the others; only the per-harmonic pairs $(k_{x,h}, k_{y,h})$ enter, through (42).

Remark 3 (Local TE/TM basis). The 2×2 block (42) admits a further, per-harmonic rotation that renders it *diagonal*. Its eigenvectors are the two transverse polarizations of the homogeneous plane wave with wavevector $(k_{x,h}, k_{y,h}, q_h)$. For the s -polarized mode the electric field is perpendicular to the incidence plane,

$$\mathbf{e}_h^{(s)} = \frac{1}{\sqrt{k_{x,h}^2 + k_{y,h}^2}} \begin{bmatrix} -k_{y,h} \\ k_{x,h} \end{bmatrix}, \quad \mathbf{K}^{(h)} \mathbf{e}_h^{(s)} = \omega^2 \mathbf{e}_h^{(s)}, \quad (46)$$

with eigenvalue ω^2 (the s -polarized transverse field sees no projection onto the longitudinal direction), while the p -polarized mode lies in the incidence plane,

$$\mathbf{e}_h^{(p)} = \frac{1}{\sqrt{k_{x,h}^2 + k_{y,h}^2}} \begin{bmatrix} k_{x,h} \\ k_{y,h} \end{bmatrix}, \quad \mathbf{K}^{(h)} \mathbf{e}_h^{(p)} = \left(\omega^2 - \frac{k_{x,h}^2 + k_{y,h}^2}{\varepsilon_0} \right) \mathbf{e}_h^{(p)} = \frac{q_h^2}{\varepsilon_0} \mathbf{e}_h^{(p)}. \quad (47)$$

In the interleaved per-harmonic basis with each harmonic rotated to $(\mathbf{e}_h^{(s)}, \mathbf{e}_h^{(p)})$, every homogeneous interface matrix becomes a *scalar* diagonal per polarization, and the recursion (48) degenerates to two independent scalar Fresnel recursions per harmonic — the classical quasi-1D diagonal core, now valid at arbitrary $(k_{x,h}, k_{y,h})$. We retain the 2×2 form in (48) because it avoids introducing a per-harmonic rotation in the assembly stage: the block-diagonal structure in the fixed component basis is what the cascade (49) multiplies against the dense prefix. The rotation is useful, however, when the suffix alone is composed repeatedly (e.g. in a wavelength scan), since it halves the work of the per-harmonic recursion.

3.2 The per-harmonic uniform-suffix algorithm

The picture in words. Because this section is the algorithmic core of the paper, we first restate the objects it manipulates in plain terms; the precise notation is introduced in section 2. The Fourier modal method represents the field inside a periodic stack as a sum of plane waves. Each plane wave is labeled by an integer h — a “harmonic” or diffraction order — and is characterized by a fixed sideways momentum $(k_{x,h}, k_{y,h})$ in the plane of the layers, plus a momentum q_h along the stacking direction z that the material fixes. We keep n such plane waves in total, so a scalar field is a list of n Fourier coefficients and a vector field is two such lists (one per transverse electric component E_x, E_y). The two numbers $(E_{x,h}, E_{y,h})$ of a single harmonic are the only two degrees of freedom that a homogeneous layer keeps coupled.

Two objects recur throughout. *Amplitudes*: at any height we split the field into waves going down the stack (forward, denoted $\mathbf{a}$) and waves going back up (backward, denoted $\mathbf{b}$); each is a list of $2n$ numbers, two transverse components per harmonic. *Scattering matrix S* : a table that answers “if these amplitudes enter a block of layers, which amplitudes come out?”. It is assembled one interface at a time by the star product (37); computing reflectivity and transmissivity is simply building S for the whole stack and applying it to the incident amplitudes.

In this vocabulary lemma 1 is one sentence of physics: *a homogeneous layer conserves sideways momentum, so the order h never changes*. A wave of order h crossing a flat interface can be reflected, can be transmitted, and can have its polarization rotated into the other polarization, but it cannot jump to another order. Each harmonic therefore lives in its own private 2×2 world — the two polarizations — and no homogeneous stack can mix one harmonic with another. Everything in this section is the practical exploitation of that single sentence.

Motivation. The generic star product (37) treats every interface as a dense $2n \times 2n$ scatterer and therefore spends cubic work at *each* of the L interfaces of a multilayer. This is hugely wasteful: in the interleaved (harmonic-major) basis of lemma 1 the scattering matrix of any all-homogeneous region is block diagonal, with one 2×2 block per harmonic, so a dense implementation is multiplying matrices whose entries are exactly zero in the right basis — it pays for n^2 coupling coefficients when the physics only ever uses the n diagonal blocks.

When the homogeneous layers outnumber the patterned ones this waste dominates. The EUV-mask geometry of Section 9 has a single patterned absorber on top of roughly 80 homogeneous interfaces; treating those interfaces as dense scatterers repeats cubic work even though they cannot couple distinct harmonics. The algorithm below replaces those 80 dense star products by $80 \times n$ independent 2×2 updates. Nothing is approximated: the reduction is a change of basis, and the resulting observables are bit-identical to the generic path.

The split. The one structural fact we use is that *only patterned layers couple distinct harmonics*. Split the stack (indexed $0, \dots, L-1$ from the incident medium) at the first layer m of its trailing all-homogeneous region, so that layers $m, \dots, L-1$ are homogeneous and every patterned layer has index $< m$. For

vacuum | absorber | cap | $40 \times (\text{Mo/Si})$ | substrate,

the absorber is the single patterned layer, m points at the cap, and the suffix is $\text{cap} + 40 \times (\text{Mo/Si}) + \text{substrate}$. The algorithm then has three steps.

1. *Prefix.* Compute $\mathbf{L} = S(0, m)$, the scattering matrix of the patterned part, with the general dense star product (37). This part is genuinely dense — the pattern does couple harmonics — but it is *short*: it involves only the few patterned layers and their immediate neighbors, so its cost is a small, fixed number of $O((2n)^3)$ steps, independent of how many homogeneous layers follow. The prefix is continued up to and including the boundary layer adjacent to the suffix, so that $\mathbf{L}$ and $\mathbf{R}$ share the boundary layer m in step 3 below.
2. *Suffix.* Compute $\mathbf{R} = S(m, L)$, the scattering matrix of the all-homogeneous suffix, by the per-harmonic recursion. By Corollary 2, $\mathbf{R}$ is block diagonal in the interleaved basis, so it suffices to compute, for each harmonic $h = 1, \dots, n$, the four 2×2 blocks $\mathbf{S}_{11}^{(h)}, \mathbf{S}_{12}^{(h)}, \mathbf{S}_{21}^{(h)}, \mathbf{S}_{22}^{(h)}$. Each recursion is the star product (37) restricted to the h -th block, and it uses only per-harmonic quantities:

- the 2×2 dispersion matrix $\mathbf{K}^{(h)}$ of (42), which encodes how the two transverse components of harmonic h propagate in a homogeneous medium of permittivity ε ;
- the propagation constant q_h (with $q_h^2 = \varepsilon\omega^2 - k_{x,h}^2 - k_{y,h}^2$), which enters through the two scalar phase factors $d_{1,h} = e^{iq_{\ell,h}t_{\ell}}$ and $d_{2,h} = e^{iq_{\ell+1,h}t_{\ell+1}}$; these advance the forward and backward amplitudes across the full thicknesses $t_{\ell}, t_{\ell+1}$ of the two layers that meet at the interface being processed;
- the per-harmonic interface matrix $\mathbf{P}^{(h)} = \frac{q_{\ell,h}}{q_{\ell+1,h}} (\mathbf{K}_{\ell}^{(h)})^{-1} \mathbf{K}_{\ell+1}^{(h)}$ of (45), together with $\mathbf{Q}^{(h)} = \mathbf{I}_2$ (because $\Phi = \mathbf{I}$ for every homogeneous layer, so $\mathbf{Q} = \Phi_{\ell}^{-1} \Phi_{\ell+1} = \mathbf{I}$). These assemble into the transmission/reflection blocks of (31),

$$\mathbf{T}_{11}^{(h)} = \frac{1}{2} (\mathbf{I}_2 + \mathbf{P}^{(h)}), \quad \mathbf{T}_{12}^{(h)} = \frac{1}{2} (\mathbf{I}_2 - \mathbf{P}^{(h)}).$$

Starting from the empty suffix — the scattering matrix of “no layers”, which transmits without reflecting, so $\mathbf{S}_{11}^{(h)} = \mathbf{S}_{22}^{(h)} = \mathbf{I}_2$ and $\mathbf{S}_{12}^{(h)} = \mathbf{S}_{21}^{(h)} = \mathbf{0}$ — sweep the interfaces $m \rightarrow m+1 \rightarrow \dots \rightarrow L-1$ and, at each, apply the 2×2 star product

$$\begin{aligned} \mathbf{M}^{(h)} &= \left(\mathbf{T}_{11}^{(h)} - d_{1,h} \mathbf{S}_{12}^{(h)} \mathbf{T}_{12}^{(h)} \right)^{-1}, \\ \mathbf{S}_{11}^{(h)} &\leftarrow \mathbf{M}^{(h)} d_{1,h} \mathbf{S}_{11}^{(h)}, \quad \mathbf{S}_{12}^{(h)} \leftarrow \mathbf{M}^{(h)} \left(d_{1,h} \mathbf{S}_{12}^{(h)} \mathbf{T}_{11}^{(h)} - \mathbf{T}_{12}^{(h)} \right) d_{2,h}, \\ \mathbf{S}_{21}^{(h)} &\leftarrow \mathbf{S}_{21}^{(h)} + \mathbf{S}_{22}^{(h)} \mathbf{T}_{12}^{(h)} \mathbf{S}_{11}^{(h)}, \quad \mathbf{S}_{22}^{(h)} \leftarrow \mathbf{S}_{22}^{(h)} \left(\mathbf{T}_{11}^{(h)} d_{2,h} + \mathbf{T}_{12}^{(h)} \mathbf{S}_{12}^{(h)} \right). \end{aligned} \quad (48)$$

Every line is a 2×2 matrix operation, so one interface costs $O(1)$ per harmonic (with a single 2×2 inversion in $\mathbf{M}^{(h)}$). The first line is the multiple-scattering resolvent: the combination $\mathbf{T}_{11}^{(h)} - d_{1,h} \mathbf{S}_{12}^{(h)} \mathbf{T}_{12}^{(h)}$ collects all repeated bounces between the new interface and the part of the stack already accumulated to its right, and inverting it sums that geometric series. The four updates then describe, for this one harmonic, how the two polarizations reflect and transmit — with possible polarization mixing — through the homogeneous multilayer accumulated so far; they are (37) with every matrix replaced by its h -th 2×2 block. After the sweep the suffix is stored as n quadruples of 2×2 blocks (never as a dense $2n \times 2n$ matrix), at a total cost of $O(n(L-m))$ operations.

3. *Assembly.* Join the dense prefix $\mathbf{L}$ and the block-diagonal suffix $\mathbf{R}$ by the overlapping cascade (49). The two matrices overlap in the sense that both reference the amplitudes (a_m, b_m) on the shared boundary layer m ; the cascade is simply the Gaussian elimination of those two internal amplitudes,

Table 1: Cost of the S -matrix of a stack with a patterned prefix and an N_{layers} -layer uniform suffix ($N = 2n$).

	generic	per-harmonic suffix
uniform interfaces	$O((2n)^3 N_{\text{layers}})$	$O(n N_{\text{layers}})$
patterned prefix	$O((2n)^3)$ per layer	same
assembly	$O((2n)^3)$	$O((2n)^3)$ (once)

exactly as in (41). Its explicit form is

$$\begin{aligned}
 \mathbf{S}_{11} &= \mathbf{R}_{11} \mathbf{M} \mathbf{L}_{11}, \\
 \mathbf{S}_{12} &= \mathbf{R}_{11} \mathbf{M} \mathbf{L}_{12} \mathbf{R}_{22} + \mathbf{R}_{12}, \\
 \mathbf{S}_{21} &= \mathbf{L}_{21} + \mathbf{L}_{22} \mathbf{R}_{21} \mathbf{M} \mathbf{L}_{11}, \\
 \mathbf{S}_{22} &= \mathbf{L}_{22} \mathbf{R}_{21} \mathbf{M} \mathbf{L}_{12} \mathbf{R}_{22} + \mathbf{L}_{22} \mathbf{R}_{22}.
 \end{aligned} \tag{49}$$

The point of keeping the suffix in block form is that the products $\mathbf{L}_{12} \mathbf{R}_{21}$ and $\mathbf{R}_{11}(\cdot)$ are $O(n)$ block scalings rather than dense products, because $\mathbf{R}_{21}, \mathbf{R}_{22}$ are 2×2 -block-diagonal. The only dense $O((2n)^3)$ operations in the whole assembly are the single inverse $(\mathbf{I} - \mathbf{L}_{12} \mathbf{R}_{21})^{-1}$ and the two dense products $\mathbf{M} \mathbf{L}_{11}$ and $\mathbf{M}(\mathbf{L}_{12} \mathbf{R}_{22})$.

A worked micro-example: one dielectric interface. To make the 2×2 blocks concrete, consider a single flat interface between two homogeneous media of refractive indices $n_\ell, n_{\ell+1}$, at normal incidence, so $k_{x,h} = k_{y,h} = 0$ and $q_h = \omega n$. Then (42) collapses to the scalar multiple $\mathbf{K}^{(h)} = \omega^2 \mathbf{I}_2$, and the per-harmonic interface matrix (45) becomes $\mathbf{P}^{(h)} = \frac{q_\ell}{q_{\ell+1}} \mathbf{I}_2 = \frac{n_\ell}{n_{\ell+1}} \mathbf{I}_2$ — a multiple of the identity, reflecting that at normal incidence the two polarizations behave identically. The transmission/reflection blocks are then

$$\mathbf{T}_{11}^{(h)} = \frac{n_\ell + n_{\ell+1}}{2n_{\ell+1}} \mathbf{I}_2, \quad \mathbf{T}_{12}^{(h)} = \frac{n_{\ell+1} - n_\ell}{2n_{\ell+1}} \mathbf{I}_2,$$

and a single star step produces, up to the reference-height phase factors $d_{1,h}, d_{2,h}$, the familiar amplitude reflection coefficient $r = \frac{n_\ell - n_{\ell+1}}{n_\ell + n_{\ell+1}}$ with no polarization conversion. The per-harmonic recursion is therefore literally “a Fresnel recursion per diffraction order”: at oblique incidence the 2×2 block stops being a scalar multiple and keeps the two distinct Fresnel problems (the s and p polarizations) correctly coupled, which is precisely what the generic dense code would compute, only now as n tiny 2×2 problems.

All quantities in steps 2–3 are understood in the interleaved basis; the prefix, computed in component-major order, is brought into this basis by the exact permutation Π of (44) — a pure relabeling of rows and columns with no arithmetic.

Derivation of the overlapping cascade. With the scattering convention $[a_{\text{out}}; b_{\text{in}}] = S[a_{\text{in}}; b_{\text{out}}]$, the prefix $\mathbf{L} = S(0, m)$ and the suffix $\mathbf{R} = S(m, L)$ both reference the amplitudes (a_m, b_m) on the shared layer m :

$$a_m = \mathbf{L}_{11} a_0 + \mathbf{L}_{12} b_m, \quad b_m = \mathbf{R}_{21} a_m + \mathbf{R}_{22} b_L, \quad a_L = \mathbf{R}_{11} a_m + \mathbf{R}_{12} b_L. \tag{50}$$

Eliminating a_m, b_m between the first two relations gives $(\mathbf{I} - \mathbf{L}_{12} \mathbf{R}_{21}) a_m = \mathbf{L}_{11} a_0 + \mathbf{L}_{12} \mathbf{R}_{22} b_L$, i.e. $a_m = \mathbf{M}(\mathbf{L}_{11} a_0 + \mathbf{L}_{12} \mathbf{R}_{22} b_L)$ with $\mathbf{M} = (\mathbf{I} - \mathbf{L}_{12} \mathbf{R}_{21})^{-1}$. Substituting into a_L yields $\mathbf{S}_{11} = \mathbf{R}_{11} \mathbf{M} \mathbf{L}_{11}$ and $\mathbf{S}_{12} = \mathbf{R}_{11} \mathbf{M} \mathbf{L}_{12} \mathbf{R}_{22} + \mathbf{R}_{12}$; the backward equations give $\mathbf{S}_{21} = \mathbf{L}_{21} + \mathbf{L}_{22} \mathbf{R}_{21} \mathbf{M} \mathbf{L}_{11}$ and $\mathbf{S}_{22} = \mathbf{L}_{22} \mathbf{R}_{21} \mathbf{M} \mathbf{L}_{12} \mathbf{R}_{22} + \mathbf{L}_{22} \mathbf{R}_{22}$, in agreement with (49).

Table 1 summarizes the scaling. For a stack whose patterned layers are few (as in the EUV-mask geometry: vacuum | absorber | cap | $40 \times (\text{Mo/Si})$ | substrate), the entire multilayer suffix is reduced from 80 dense $O((2n)^3)$ star products to $O(n \cdot 82)$ scalar- 2×2 operations, and the residual dense work is one prefix and one cascade.

In short: the algorithm is (i) a dense, exact treatment of the few layers that actually mix harmonics, (ii) an independent 2×2 Fresnel-style recursion per harmonic through the many layers that do not, and (iii) a single elimination step that glues the two pieces at their shared boundary. Steps (i) and (iii) are fixed one-time costs; step (ii) scales linearly in both the harmonic count and the number of uniform layers, which is what removes the cubic cost of a multilayer.

3.3 Relation to quasi-1D and periodic-stack methods

When the harmonic set is the x -only row and $k_{y,h} = 0$ for all h , the blocks (42) are diagonal and (48) reduces to independent *scalar* Fresnel-like recursions per harmonic — the classical quasi-1D trick. When the suffix is periodic (a repeated bilayer), the per-harmonic recursion handles it as cheaply as the periodic cores exponentiated by the Redheffer cascade; no periodicity assumption is required. The factorization therefore subsumes both the quasi-1D diagonal core and periodic binary exponentiation as special cases.

4 Exact decompositions of the patterned-layer eigenproblem

What this section does, in words. A patterned layer couples different harmonics, so its operator (23) is not block diagonal, and its spectrum must in general be found by a dense eigensolve of size $2n$. The eigenvalues q_j^2 are the squared vertical wavevectors of the layer’s *modes* — the standing or running waves that travel through the layer without changing their transverse shape — and the eigenvectors store those shapes. This section shows that for a y -invariant grating the $2n \times 2n$ problem collapses to two $n \times n$ problems, plus an explicit recipe for the eigenvectors. The reason, stated physically, is simple: at normal incidence the two polarizations do not mix (a block-diagonal matrix), and at oblique incidence they mix only “one way” (a block-triangular matrix). Both cases drop the expensive diagonalization by a factor four; the second still needs a small extra step to rebuild the coupled eigenvectors, which Section 4.3 gives in closed form.

The remaining $O((2n)^3)$ bottleneck is the eigenproblem (23) of the patterned layer. We show that for the two most common structures it decouples exactly into two $n \times n$ problems, at $4\times$ fewer eigensolver flops, with a constructive formula for the eigenvectors. Throughout, the harmonic set is the 1D row of a y -invariant grating, so that $\mathbf{k}_y = \kappa_y \mathbf{1}$ is a constant vector, but we emphasize that the block-diagonal case also covers certain 2D patterns (see Section 10).

The following distinction is useful. The homogeneous-layer reduction follows from the absence of lateral material modulation and is valid for every harmonic set. The patterned-layer reduction instead requires a special relation between the material convolution operator and the transverse wavevector. In a y -invariant grating, the retained harmonics have the form $\mathbf{G}_h = (m_h, 0)$, so

$$K_y = \kappa_y I_n, \quad K_x = D_x = \text{diag}(k_{x,1}, \dots, k_{x,n}). \quad (51)$$

The scalar κ_y need not vanish: it is nonzero for conical or oblique illumination. This is precisely why the oblique case is not TE/TM separated, but still has a triangular spectral structure.

4.1 Block-diagonal modal operator at normal incidence

At normal incidence, $k_{x0} = k_{y0} = 0$. For a y -invariant profile one also has $\mathbf{k}_y = \mathbf{0}$. Thus the block in (25) reduces to

$$\mathbf{J}_k = \begin{bmatrix} \mathbf{0} \\ \text{diag}(\mathbf{k}_x) \end{bmatrix}.$$

Writing $D_x = \text{diag}(\mathbf{k}_x)$ gives

$$\mathbf{J}_k \mathbf{E} = \begin{bmatrix} \mathbf{0} \\ D_x \mathbf{E} \end{bmatrix}, \quad (\mathbf{J}_k \mathbf{E}) \mathbf{J}_k^\top = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & D_x \mathbf{E} D_x \end{bmatrix}, \quad (52)$$

so that $\mathbf{K} = \text{diag}(\omega^2 \mathbf{I}, \omega^2 \mathbf{I} - D_x E D_x)$. Similarly $\mathbf{k} = [D_x; \mathbf{0}]$ gives $\mathbf{k} \mathbf{k}^\top = \text{diag}(D_x^2, \mathbf{0})$, and $\text{ep2} = \text{blkdiag}(C, C)$. Hence

$$\mathbf{M} = \text{ep2} \mathbf{K} - \mathbf{k} \mathbf{k}^\top = \begin{bmatrix} C \omega^2 - D_x^2 & \mathbf{0} \\ \mathbf{0} & C(\omega^2 \mathbf{I} - D_x E D_x) \end{bmatrix} \equiv \begin{bmatrix} \mathbf{M}_{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_{22} \end{bmatrix}, \quad (53)$$

with, using $CE = \mathbf{I}$,

$$\mathbf{M}_{11} = \omega^2 C - D_x^2, \quad \mathbf{M}_{22} = \omega^2 C - C D_x C^{-1} D_x. \quad (54)$$

The $2n \times 2n$ eigenproblem (23) therefore decouples exactly into two $n \times n$ problems, $\text{eig}(\mathbf{M}) = \text{eig}(\mathbf{M}_{11}) \cup \text{eig}(\mathbf{M}_{22})$, $\Phi = \text{blkdiag}(\Phi_1, \Phi_2)$, reducing the eigensolver flops by a factor four. (For a D4-symmetric 2D pattern at normal incidence the off-diagonal blocks also vanish exactly, and the same decomposition applies.)

Remark 4 (Polarization interpretation of the two blocks). The decoupling (53) is the Fourier-space form of the TE/TM separation of a one-dimensional grating. For $\varepsilon = \varepsilon(x)$ the two polarizations are distinguished by their behavior at the material interfaces. One polarization has its transverse electric field parallel to the grooves and hence continuous across the pattern; it is governed by the Laurent convolution C alone, which is exactly the block $\mathbf{M}_{11} = \omega^2 C - D_x^2$ of (54). The orthogonal polarization has a normal electric component that jumps at the interfaces; it therefore carries the inverse convolution $E = C^{-1}$ in the form $\mathbf{M}_{22} = \omega^2 C - C D_x C^{-1} D_x$, which is Li's inverse factorization rule [7] for the discontinuous normal field. The two blocks are precisely the two standard one-dimensional polarization operators, and their separation is what Section 5 exploits to run the entire scattering recursion per polarization.

4.2 Block-lower-triangular modal operator at oblique incidence

At oblique incidence $k_{y0} \neq 0$ the 1D harmonic set still has $\mathbf{k}_y = \kappa_y \mathbf{1}$ (a constant vector), so that $D_y = \text{diag}(\mathbf{k}_y) = \kappa_y \mathbf{I}$ and $\mathbf{J}_k = [-\kappa_y \mathbf{I}; D_x]$. Evaluating (25)–(24) blockwise,

$$\mathbf{J}_k E = \begin{bmatrix} -\kappa_y E \\ D_x E \end{bmatrix}, \quad (\mathbf{J}_k E) \mathbf{J}_k^\top = \begin{bmatrix} \kappa_y^2 E & -\kappa_y E D_x \\ -\kappa_y D_x E & D_x E D_x \end{bmatrix}, \quad (55)$$

so that

$$\mathbf{K} = \begin{bmatrix} \omega^2 \mathbf{I} - \kappa_y^2 E & \kappa_y E D_x \\ \kappa_y D_x E & \omega^2 \mathbf{I} - D_x E D_x \end{bmatrix}, \quad \mathbf{k} \mathbf{k}^\top = \begin{bmatrix} D_x^2 & \kappa_y D_x \\ \kappa_y D_x & \kappa_y^2 \mathbf{I} \end{bmatrix}. \quad (56)$$

The four blocks of $\mathbf{M} = \text{ep2} \mathbf{K} - \mathbf{k} \mathbf{k}^\top$ are

$$\mathbf{M}_{11} = C(\omega^2 \mathbf{I} - \kappa_y^2 E) - D_x^2 = \omega^2 C - \kappa_y^2 \mathbf{I} - D_x^2, \quad (57)$$

$$\mathbf{M}_{12} = C \kappa_y E D_x - \kappa_y D_x = \kappa_y (CE - \mathbf{I}) D_x = \mathbf{0}, \quad (58)$$

$$\mathbf{M}_{21} = C \kappa_y D_x E - \kappa_y D_x = \kappa_y (C D_x C^{-1} - D_x) \neq \mathbf{0}, \quad (59)$$

$$\mathbf{M}_{22} = C(\omega^2 \mathbf{I} - D_x E D_x) - \kappa_y^2 \mathbf{I} = \omega^2 C - \kappa_y^2 \mathbf{I} - C D_x C^{-1} D_x. \quad (60)$$

Because $\mathbf{M}_{12} = \mathbf{0}$, $\mathbf{M}$ is block-lower-triangular,

$$\mathbf{M} = \begin{bmatrix} \mathbf{M}_{11} & \mathbf{0} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{bmatrix}, \quad (61)$$

and $\det(\mathbf{M} - \lambda \mathbf{I}) = \det(\mathbf{M}_{11} - \lambda \mathbf{I}) \det(\mathbf{M}_{22} - \lambda \mathbf{I})$: the eigenvalues split exactly into two n -dimensional problems $\text{eig}(\mathbf{M}) = \text{eig}(\mathbf{M}_{11}) \cup \text{eig}(\mathbf{M}_{22})$, a $4\times$ reduction in eigensolver flops. The eigenvectors, however, remain coupled through the nonzero lower-left block $\mathbf{M}_{21}$; their exact reconstruction is the subject of the next subsection.

4.3 Exact reconstruction of the coupled eigenvectors

In words: once the two diagonal blocks are diagonalized separately, the only missing information is the cross-talk between the two polarizations stored in the lower-left block $\mathbf{M}_{21}$. That cross-talk is recovered by solving one linear (Sylvester) equation, whose solution is written down explicitly when the two spectra do not overlap. Because $\mathbf{M}$ is block-lower-triangular, the eigenvectors of the two eigenvalue families are constructed as follows.

This construction is an instance of block diagonalization. Suppose $\mathbf{M}_{11}$ and $\mathbf{M}_{22}$ are diagonalizable and their spectra do not intersect. Seek a similarity transformation

$$\mathbf{X} = \begin{bmatrix} I & 0 \\ Y & I \end{bmatrix} \quad (62)$$

that removes the lower-left block. Direct multiplication gives

$$\mathbf{X}^{-1}\mathbf{M}\mathbf{X} = \begin{bmatrix} \mathbf{M}_{11} & 0 \\ \mathbf{M}_{21} + \mathbf{M}_{22}Y - Y\mathbf{M}_{11} & \mathbf{M}_{22} \end{bmatrix}. \quad (63)$$

Thus Y must solve the Sylvester equation

$$\mathbf{M}_{22}Y - Y\mathbf{M}_{11} = -\mathbf{M}_{21}. \quad (64)$$

The spectral decompositions $\mathbf{M}_{11} = \Phi_A \Lambda_A \Phi_A^{-1}$ and $\mathbf{M}_{22} = \Phi_C \Lambda_C \Phi_C^{-1}$ convert (64) into

$$\Lambda_C Z - Z\Lambda_A = -\Phi_C^{-1}\mathbf{M}_{21}\Phi_A, \quad Z = \Phi_C^{-1}Y\Phi_A. \quad (65)$$

Since the two diagonal matrices act by row and column scaling,

$$Z_{ij} = -\frac{(\Phi_C^{-1}\mathbf{M}_{21}\Phi_A)_{ij}}{\lambda_{C,i} - \lambda_{A,j}}. \quad (66)$$

This explains both the reconstruction formula and its failure mode: when an A and C eigenvalue approach one another, the Sylvester operator becomes ill-conditioned and the full $2n$ eigensolve is the appropriate fallback.

Columns for $\lambda_{A,j} = \text{eig}_j(\mathbf{M}_{11})$. Let $\mathbf{M}_{11} = \Phi_A \Lambda_A \Phi_A^{-1}$ with columns $\Phi_A = [\mathbf{a}_1, \dots, \mathbf{a}_n]$ and $\mathbf{M}_{11}\mathbf{a}_j = \lambda_{A,j}\mathbf{a}_j$, and let $\mathbf{M}_{22} = \Phi_C \Lambda_C \Phi_C^{-1}$. Writing an eigenvector as $[\mathbf{a}_j; \mathbf{v}_j]$, the second block row of $(\mathbf{M} - \lambda_{A,j}\mathbf{I})[\mathbf{a}_j; \mathbf{v}_j] = 0$ reads

$$(\mathbf{M}_{22} - \lambda_{A,j}\mathbf{I})\mathbf{v}_j = -\mathbf{M}_{21}\mathbf{a}_j, \quad (67)$$

the first block row being satisfied identically. Equation (67) is a linear system for $\mathbf{v}_j$ with the matrix $\mathbf{M}_{22} - \lambda_{A,j}\mathbf{I}$. When $\lambda_{A,j} \notin \text{eig}(\mathbf{M}_{22})$ this matrix is nonsingular and the resolvent expansion in the eigenbasis of $\mathbf{M}_{22}$ applies: with $\mathbf{M}_{22} = \Phi_C \Lambda_C \Phi_C^{-1}$,

$$(\mathbf{M}_{22} - \lambda_{A,j}\mathbf{I})^{-1} = \Phi_C (\Lambda_C - \lambda_{A,j}\mathbf{I})^{-1}\Phi_C^{-1} = \Phi_C \text{diag}\left(\frac{1}{\lambda_{C,i} - \lambda_{A,j}}\right)\Phi_C^{-1}, \quad (68)$$

so that

$$\mathbf{v}_j = -(\mathbf{M}_{22} - \lambda_{A,j}\mathbf{I})^{-1}\mathbf{M}_{21}\mathbf{a}_j = -\Phi_C \text{diag}\left(\frac{1}{\lambda_{C,i} - \lambda_{A,j}}\right)\Phi_C^{-1}\mathbf{M}_{21}\mathbf{a}_j. \quad (69)$$

Columns for $\lambda_{C,j} = \text{eig}_j(\mathbf{M}_{22})$. Writing an eigenvector as $[\mathbf{u}_j; \mathbf{c}_j]$, the second block row $(\mathbf{M}_{22} - \lambda_{C,j}\mathbf{I})\mathbf{c}_j = 0$ is satisfied by $\mathbf{c}_j = \Phi_C[:, j]$; the first block row $(\mathbf{M}_{11} - \lambda_{C,j}\mathbf{I})\mathbf{u}_j = \mathbf{0}$ then forces $\mathbf{u}_j = \mathbf{0}$ provided $\lambda_{C,j} \notin \text{eig}(\mathbf{M}_{11})$. The corresponding modal columns are therefore $[\mathbf{0}; \Phi_C[:, j]]$.

Collecting both families gives the block-lower-triangular modal matrix

$$\Phi = \begin{bmatrix} \Phi_A & \mathbf{0} \\ \mathbf{V} & \Phi_C \end{bmatrix}, \quad \mathbf{V} = -\Phi_C \left((\Phi_C^{-1} \mathbf{M}_{21} \Phi_A) \oslash \mathbf{D} \right), \quad \mathbf{D}_{ij} = \lambda_{C,i} - \lambda_{A,j}, \quad (70)$$

where $\oslash$ denotes element-wise division. If $\mathbf{X} = \Phi_C^{-1} \mathbf{M}_{21} \Phi_A$, then the (i, j) entry of the Hadamard quotient is

$$(\mathbf{X} \oslash \mathbf{D})_{ij} = \frac{X_{ij}}{\lambda_{C,i} - \lambda_{A,j}},$$

which agrees column by column with (69). The reconstruction (70) costs two $n \times n$ eigendecompositions, one $n \times n$ solve $(\Phi_C^{-1} \mathbf{M}_{21} \Phi_A)$, and two $O(n^3)$ products, versus one $2n \times 2n$ eigendecomposition; the results are numerically identical to the full solve (verified to 17 digits in the scattering observables). If a pair of eigenvalues becomes exactly degenerate ($\min_{ij} |\mathbf{D}_{ij}| \rightarrow 0$) the resolvent (69) is singular and the code falls back to the full $2n \times 2n$ solve; the detection is exact (off-diagonal-block norm test), so the result is never silently wrong.

5 Polarization-decoupled scattering at normal incidence

This short section spells out the payoff of Section 4.1: at normal incidence the two polarizations are independent through the *whole* stack, not just inside each layer. In plain terms, TE- and TM-polarized light at normal incidence pass through the structure without exchanging energy, so the entire scattering computation — not only the eigenvalue problem — runs as two separate problems of half the size. The section below exploits this to quarter the remaining dense work.

At normal incidence with block-diagonal $\mathbf{K}$, Φ , $\mathbf{q}$ (Section 4.1), every interface matrix (29) is block-diagonal with two $n \times n$ blocks (the TE and TM polarizations). The Redheffer recursion and the cascade then run independently per polarization on $n \times n$ blocks — a further $4\times$ reduction on the (already reduced) patterned prefix and the assembly — and the result is reassembled as $\mathbf{S} = \text{blkdiag}(\mathbf{S}_{\text{TE}}, \mathbf{S}_{\text{TM}})$. Combined with the per-harmonic suffix of Section 3.2, the quasi-1D normal-incidence solve is thus scalar per harmonic throughout the uniform suffix and $n \times n$ -block in the prefix.

6 Binary gratings: low-rank permittivity structure

A “binary” grating is one whose permittivity takes only two values — a step profile, such as a metal bar or a rectangular hole on a substrate. Such profiles pervade lithography, and they have a feature that generic smooth profiles lack: their Fourier coefficients are known in closed form, and the matrix built from those coefficients is very nearly low rank. This section records both facts. The first (Sections 6.1 and its discussion) replaces grid sampling with exact coefficients and removes the pixelation error of a discrete grid; the second (Section 6.3) observes a diagonal-plus-low-rank form of the patterned operator that points toward eigensolvers whose cost tracks a small rank rather than n , though that path is left to future work.

6.1 Fourier coefficients of a step profile

For the step-rect profile $\varepsilon(x) = \varepsilon_1$ on $|x| < x_1$, ε_2 otherwise, the Fourier coefficients follow from a one-line integral. Writing $\varepsilon(x) = \varepsilon_2 + \Delta\varepsilon \chi(x)$ with $\Delta\varepsilon = \varepsilon_1 - \varepsilon_2$ and χ the indicator of $|x| < x_1$, the coefficient $\hat{\varepsilon}_m = (1/L_x) \int_{-L_x/2}^{L_x/2} \varepsilon(x) e^{-i\beta x} dx$ with $\beta = 2\pi m/L_x$ splits into the constant part and the pattern part:

$$\hat{\varepsilon}_m = \varepsilon_2 \delta_{m0} + \frac{\Delta\varepsilon}{L_x} \int_{-x_1}^{x_1} e^{-i\beta x} dx. \quad (71)$$

For $m = 0$ the integral gives $2x_1$, so $\hat{\varepsilon}_0 = \varepsilon_2 + \Delta\varepsilon f$ with the fill fraction $f = 2x_1/L_x$. For $m \neq 0$ the constant part integrates to zero, and

$$\frac{1}{L_x} \int_{-x_1}^{x_1} e^{-i\beta x} dx = \frac{1}{L_x} \frac{e^{-i\beta x_1} - e^{i\beta x_1}}{-i\beta} = \frac{1}{L_x} \frac{2 \sin(\beta x_1)}{\beta} = \frac{2x_1}{L_x} \frac{\sin(\pi m f)}{\pi m f} = f \operatorname{sinc}(mf), \quad (72)$$

where $\operatorname{sinc}(z) = \sin(\pi z)/(\pi z)$. Hence

$$\hat{\varepsilon}_m = \begin{cases} \varepsilon_2 + \Delta\varepsilon f, & m = 0, \\ \Delta\varepsilon f \operatorname{sinc}(mf), & m \neq 0. \end{cases} \quad (73)$$

The Toeplitz convolution matrix is therefore $C = \varepsilon_2 \mathbf{I} + \Delta\varepsilon f \mathbf{S}$ with the “discrete sinc” matrix $\mathbf{S}_{ij} = \operatorname{sinc}(f(i - j))$.

6.2 The Slepian (prolate) matrix

The matrix $\mathbf{S}$ is the finite-dimensional prolate (Slepian) matrix [10]: its symbol is the indicator of the frequency band $|\theta| \leq \pi f$, and its eigenvalues obey the two-interval concentration law — approximately nf eigenvalues near 1 and the rest exponentially small,

$$\operatorname{rank}_\tau(\mathbf{S}) \approx nf + O(\log \tau^{-1}). \quad (74)$$

For the EUV example of Section 9 ($f = 500/3500 \approx 0.143$, $n = 253$) this gives a numerical rank of ≈ 52 at $\tau = 10^{-14}$, i.e. its numerical rank is approximately five times smaller than its dimension; a rank-52 reconstruction reproduces $\mathbf{S}$ to $\sim 1.5 \times 10^{-15}$. Two consequences follow.

First, *the analytic coefficients* (73) replace the grid sampling and fast Fourier transform in the construction of C , and remove the pixelation error of a discrete grid (a 1000-pixel cell quantizes the fill to $f = 0.1420$ against the exact $f = 0.142857$). For smooth profiles (Gaussian, Lorentzian) the coefficients decay exponentially and the harmonic count required for a fixed accuracy drops correspondingly.

6.3 Diagonal-plus-low-rank structure of the patterned operator

Second, *both patterned-layer eigenproblems admit diagonal-plus-low-rank representations*. Let $\mathbf{S} = U\Sigma U^\top$ be the (numerically) rank- r eigendecomposition, $U \in \mathbb{C}^{n \times r}$, $\Sigma \in \mathbb{R}^{r \times r}$ (so $U^\top U = \mathbf{I}_r$). The inverse of $C = \varepsilon_2 \mathbf{I} + \Delta\varepsilon f U\Sigma U^\top$ follows from the Woodbury identity $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$ with $A = \varepsilon_2 \mathbf{I}$, $C = \Delta\varepsilon f \Sigma$, $V = U^\top$:

$$C^{-1} = (\varepsilon_2 \mathbf{I} + \Delta\varepsilon f U\Sigma U^\top)^{-1} = \frac{1}{\varepsilon_2} \left(\mathbf{I} - U(\varepsilon_2 \mathbf{I} + \Delta\varepsilon f \Sigma)^{-1} (\Delta\varepsilon f \Sigma) U^\top \right), \quad (75)$$

where the last step uses that $(\Delta\varepsilon f \Sigma)(\varepsilon_2 \mathbf{I} + \Delta\varepsilon f \Sigma)^{-1}$ and $(\varepsilon_2 \mathbf{I} + \Delta\varepsilon f \Sigma)^{-1}(\Delta\varepsilon f \Sigma)$ commute (both are functions of Σ). Equation (75) is a diagonal plus a rank- r update, with the abbreviation $\Gamma = (\varepsilon_2 \mathbf{I} + \Delta\varepsilon f \Sigma)^{-1} \Delta\varepsilon f \Sigma$. Expanding $CDC^{-1}D$ with $D = \operatorname{diag}(\mathbf{k}_x)$ and $C = \varepsilon_2 \mathbf{I} + \Delta\varepsilon f \mathbf{S}$,

$$CDC^{-1}D = (\varepsilon_2 \mathbf{I} + \Delta\varepsilon f \mathbf{S}) D \frac{1}{\varepsilon_2} (\mathbf{I} - U\Gamma U^\top) D = D^2 + \frac{1}{\varepsilon_2} \left(\Delta\varepsilon f \mathbf{S} D^2 - \varepsilon_2 D U\Gamma U^\top D - \Delta\varepsilon f \mathbf{S} D U\Gamma U^\top D \right), \quad (76)$$

each bracketed term having rank at most r (a product of the rank- r factor $U\Gamma U^\top$ or $U\Sigma U^\top$ with diagonal matrices). Hence the patterned blocks (54) take the form

$$\mathbf{M}_{11} = \operatorname{diag}(\beta_h) + \omega^2 \Delta\varepsilon f \mathbf{S}, \quad \mathbf{M}_{22} = \operatorname{diag}(\beta'_h) + \text{rank-} \leq 4r \text{ update}, \quad (77)$$

with $\beta_h = \omega^2 \varepsilon_2 - \kappa_y^2 - k_{x,h}^2$. Equation (77) suggests matrix–vector products, subspace iteration, or secular-equation methods whose cost depends on the numerical rank r . However, the matrices in (54) are generally

non-normal, and the existence of a low-rank update alone does not imply an $O(nr + r^3)$ algorithm for their complete spectrum. We therefore treat (77) as an operator-structure observation and do not include it in the exact speedup claims below. A stable reduced-rank eigensolver, especially for complex lossy media, is left for future work.

7 Block-wise post-processing

Once the scattering matrix and the layer modes are known, the observables of physical interest — absorbed power, the Maxwell stress tensor, near-field intensities — are quadratic in the modal amplitudes. A direct evaluation materializes very large matrices ($4n \times 4n$ and $3n \times 3n$) that are mostly redundant, and this section shows that the redundancy is *exact*: the block structure of the field and of the modal z -integrals collapses every such quadratic form onto $2n \times 2n$ blocks, cutting the transient memory by roughly an order of magnitude with no change in the result.

The absorbed power in a layer is $\text{val} = \text{tr}(\mathbf{abM}\mathbf{F}^\dagger\mathbf{M}_{\text{tot}}\mathbf{F})$, where $\mathbf{ab} = [\mathbf{a}_i; \mathbf{b}_i] \in \mathbb{C}^{4n}$ stacks the forward and backward modal amplitudes, $\mathbf{abM} = \mathbf{ab}\mathbf{ab}^\dagger \odot \mathbf{M}_t$ is a Hadamard (element-wise) product with the modal z -integral matrix $\mathbf{M}_t$, and $\mathbf{M}_{\text{tot}}$ is the block-diagonal material weighting. A direct evaluation materializes the $4n \times 4n$ matrices $\mathbf{M}_t$, $\mathbf{abM}$ and the $3n \times 3n$ matrices $\mathbf{M}_{\text{tot}}$, $\mathbf{F}$.

The block structure comes from the physics, and we make it explicit. The three Cartesian field components have $2n$ (transverse) $+n$ (longitudinal) Fourier coefficients, so $\mathbf{F}$ maps the $4n$ amplitudes to the $3n$ fields: $\mathbf{F}_{xy}$ ($2n \times 2n$) produces (E_x, E_y) from the (a, b) pair and $\mathbf{F}_{az}$ ($n \times 2n$) produces E_z . Because the transverse field (26) is antisymmetric under $a \leftrightarrow b$ ($\mathbf{E}_t \propto \mathbf{a} - \mathbf{b}$) while the longitudinal field is symmetric (E_z depends on $h_x, h_y \propto \mathbf{a} + \mathbf{b}$), the transverse/backward combinations appear with opposite signs,

$$\mathbf{F} = \begin{bmatrix} \mathbf{F}_{xy} & -\mathbf{F}_{xy} \\ \mathbf{F}_{az} & \mathbf{F}_{az} \end{bmatrix}, \quad (78)$$

where the two columns act on $\mathbf{a}$ and $\mathbf{b}$ respectively. The modal z -integral matrix $\mathbf{M}_t$ is built from the four products of a forward (e^{+iqz}) and a backward (e^{-iqz}) mode integrated over the layer thickness; these four products give only two distinct $2n \times 2n$ blocks, the “co-propagating” (a - a and b - b) block $\mathbf{M}_{aa}$ and the “counter-propagating” (a - b) block $\mathbf{M}_{ab}$,

$$\mathbf{M}_t = \begin{bmatrix} \mathbf{M}_{aa} & \mathbf{M}_{ab} \\ \mathbf{M}_{ab} & \mathbf{M}_{aa} \end{bmatrix}, \quad (79)$$

with entries

$$(\mathbf{M}_{aa})_{ij} = \int_0^t e^{i(q_j - \bar{q}_i)z} dz = \frac{e^{i(q_j - \bar{q}_i)t} - 1}{i(q_j - \bar{q}_i)}, \quad (\mathbf{M}_{ab})_{ij} = \int_0^t e^{iq_j z} e^{i\bar{q}_i z} dz = \frac{e^{iq_j t} - e^{-i\bar{q}_i t}}{i(q_j + \bar{q}_i)}, \quad (80)$$

(a small diagonal regularization $\delta = 10^{-12}$ is added to $q_j - \bar{q}_i$ so that the diagonal of $\mathbf{M}_{aa}$ is finite). Both (78) and (79) are exact; they are the only structural facts needed below.

Using the block forms (78)–(79), the interior product collapses onto $2n \times 2n$ blocks. Writing $\mathbf{T} = \mathbf{F}^\dagger \mathbf{M}_{\text{tot}} \mathbf{F}$ with $\mathbf{M}_{\text{tot}} = \text{blkdiag}(\mathbf{M}_{xy}, \mathbf{M}_z)$,

$$\mathbf{T} = \begin{bmatrix} \mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az} & -\mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az} \\ -\mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az} & \mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az} \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{bmatrix}, \quad (81)$$

with $\mathbf{A} = \mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az}$ and $\mathbf{B} = -\mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az}$. The trace is then expanded without ever forming $\mathbf{abM}$. Writing out the trace entry by entry,

$$\text{val} = \text{tr}((\mathbf{ab}\mathbf{ab}^\dagger \odot \mathbf{M}_t) \mathbf{T}) = \sum_{i,j} (\text{ab}_i \bar{\text{ab}}_j \mathbf{M}_{t,ij}) T_{ji} = \sum_{i,j} \text{ab}_i (\mathbf{M}_t \odot \mathbf{T}^\top)_{ij} \bar{\text{ab}}_j = \text{ab}^\top (\mathbf{M}_t \odot \mathbf{T}^\top) \bar{\text{ab}}, \quad (82)$$

using $(\mathbf{X} \odot \mathbf{Y})_{ij} = \mathbf{X}_{ij} \mathbf{Y}_{ij}$ and $(\mathbf{T}^\top)_{ij} = T_{ji}$. Both $\mathbf{M}_t$ and $\mathbf{T}$ share the $\{\mathbf{A}, \mathbf{B}\}$ block pattern, so $\mathbf{M}_t \odot \mathbf{T}^\top$ has the pattern $\{\mathbf{M}_{aa} \odot \mathbf{A}^\top, \mathbf{M}_{ab} \odot \mathbf{B}^\top\}$. Blockwise, with $\mathbf{W}_A = \mathbf{M}_{aa} \circ \mathbf{A}^\top$ and $\mathbf{W}_B = \mathbf{M}_{ab} \circ \mathbf{B}^\top$, and writing $\mathbf{a} = \mathbf{ab}[:2n]$, $\mathbf{b} = \mathbf{ab}[2n:]$, the quadratic form (82) splits into the four pairwise combinations

$$\text{val} = \mathbf{a}^\top \mathbf{W}_A \bar{\mathbf{a}} + \mathbf{b}^\top \mathbf{W}_A \bar{\mathbf{b}} + \mathbf{a}^\top \mathbf{W}_B \bar{\mathbf{b}} + \mathbf{b}^\top \mathbf{W}_B \bar{\mathbf{a}}. \quad (83)$$

The evaluation therefore needs two $2n \times 2n$ Hadamard products ($\mathbf{W}_A, \mathbf{W}_B$), the four block products that form $\mathbf{A}, \mathbf{B}$ in (81), and four matvecs — never materializing the $4n \times 4n$ and $3n \times 3n$ matrices. This is a $\sim 20\times$ reduction in transient memory for the absorption and stress-tensor integrals, at no change in the result (identical to the original formula to $\sim 10^{-15}$).

8 Implementation

The reductions above are implemented in a C++17 library using Eigen for the small-matrix algebra, FFTW for the permittivity Fourier transforms, and LAPACK/BLAS (OpenBLAS by default, Intel MKL optionally) for the dense non-Hermitian eigendecomposition (zggev), the factorizations (zgetrf), and the matrix products (zgemm). The library mirrors the grwa public interface and is additionally exposed through nanobind Python bindings, so that a solver invocation is a drop-in replacement for the reference code. All reductions below are structural: they are selected from exactly computable algebraic conditions on the discretized operator, and every branch has a full dense fallback, so a correct result is guaranteed even when none of the reduced forms applies.

8.1 Algorithmic workflow

The implementation follows the mathematical factorization rather than treating all layers uniformly. The workflow is:

1. construct the reciprocal lattice ($\mathbf{L}_1, \mathbf{L}_2 \rightarrow \mathbf{K}_1, \mathbf{K}_2$) and select the retained harmonic set by the circular (degeneracy-trimmed) or parallelogramic rule, so that $n_{\text{out}} \leq n_{\text{req}}$ and partial degenerate shells are never split asymmetrically;
2. form the permittivity convolution matrix $\mathbf{C}$ (and its inverse $\mathbf{E}$) only for patterned layers, via the FFT $\hat{\varepsilon}_{\mathbf{G}_i - \mathbf{G}_j}$ gathered from the transform of the sampled $\varepsilon(x, y)$ grid;
3. classify each patterned modal operator $\mathbf{M}$ by its computed off-diagonal block norms (Section 8.2): block diagonal, block lower triangular, or general;
4. solve the smallest valid eigenproblem and reconstruct the modal basis $(\Phi, \mathbf{q})$ — two $n \times n$ problems when block structured, the full $2n \times 2n$ problem otherwise;
5. locate the trailing homogeneous suffix and evaluate its scattering matrix by independent 2×2 recursions (Section 3.2);
6. compose the patterned prefix with the compact suffix using the overlapping elimination in (41), whose only dense operations are a single inverse and two products;
7. reconstruct fields and quadratic observables from the modal amplitudes, with the block-wise post-processing of Section 7.

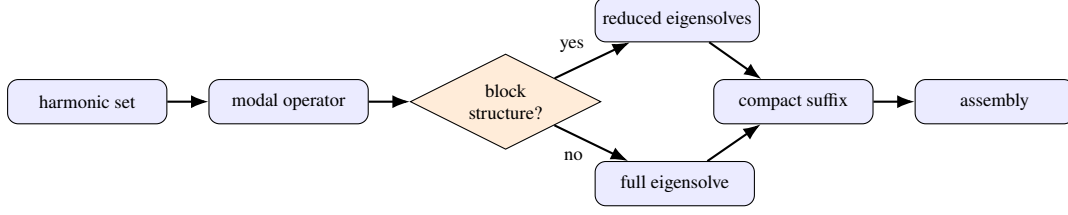


Figure 3: Structure-aware computational workflow. The branch decision is made from algebraic conditions on the discretized operator; if a condition is not met, the method falls back to the general formulation.

8.2 Structure detection and fallback

The reduced eigenproblems of Section 4 are selected by inspecting the *computed* matrix $\mathbf{M}$, not by querying the geometry. Let $\mathbf{M}_{12}, \mathbf{M}_{21}$ denote the two $n \times n$ off-diagonal blocks in the component-major ordering. The decision rule is

$$\begin{cases} \|\mathbf{M}_{12}\| = \|\mathbf{M}_{21}\| = 0 & \text{block diagonal (section 4.1),} \\ \|\mathbf{M}_{12}\| = 0, \|\mathbf{M}_{21}\| > 0 & \text{block lower triangular (section 4.2),} \\ \text{otherwise} & \text{general } 2n \times 2n \text{ problem,} \end{cases} \quad (84)$$

with the block norms measured relative to $\|\mathbf{M}\|$ and compared against a threshold of $\sim 10^{-10}$. This has two consequences. First, the branch is *safe*: for a genuinely two-dimensional pattern the off-diagonal block is $O(1)$ and the general solve is selected, so a reduction is never applied where it is not exact. Second, the branch is *robust*: it fires on the exact algebraic condition (e.g. $\mathbf{k}_y = \kappa_y \mathbf{1}$ together with isotropy $\mathbf{CE} = \mathbf{I}$ forces $\mathbf{M}_{12} = \mathbf{0}$ in exact arithmetic), independent of how the geometry was specified. The one place the numerical threshold matters is the eigenvector reconstruction of Section 4.3: when $\min_{ij} |\lambda_{C,i} - \lambda_{A,j}|$ falls below tolerance the Sylvester resolvent is ill-conditioned, and the code reverts to the full solve even though the eigenvalues alone could be split.

8.3 Engineering details

The reductions themselves determine the asymptotics; the following implementation choices realize them at measured wall time.

- *Parallel eigenpairs.* The two independent $n \times n$ eigenproblems of Sections 4.1–4.2 are solved concurrently on two threads. OpenBLAS executes `zgeev` single-threaded (LAPACK-serial), so the pair runs ~ 1.5 – $1.8\times$ faster with bit-identical results; with a threaded LAPACK (e.g. MKL) the parallelization is disabled by a flag.
- *Star-product details.* The phase factors $\mathbf{d}_1, \mathbf{d}_2$ are kept as vectors and applied as diagonal scalings ($O(n)$ per product); matrix inversions in the star update are replaced by one *LU* factorization plus two back-substitutions; the products $\mathbf{d}_1 \mathbf{S}_{12}$ and $\mathbf{S}_{22} \mathbf{T}_{12}$ are hoisted as common subexpressions; and the LAPACK workspace is reused thread-locally. Combined, these remove the three $O(n^3)$ matrix products that the naive diagonal-matrix formulation would incur per interface.
- *Uniform-layer deduplication.* A uniform layer’s $(\mathbf{K}, \mathbf{q}, \Phi)$ depends only on its permittivity and the global $(\omega, \mathbf{k}_x, \mathbf{k}_y)$, so identical- ε layers share one object, and every uniform layer points at a single shared identity Φ (since $\Phi = \mathbf{I}_{2n}$ exactly for a uniform layer). A periodic Mo/Si stack keeps one matrix per distinct material instead of one per layer — a $\sim 20\times$ persistent-storage reduction.
- *Caching.* FFTW plans are cached per grid size (with FFTW_ESTIMATE planning); identical uniform–uniform interface matrices are reused (for periodic stacks $\mathbf{Q} = \mathbf{I}$, so no per-pair inversion is needed);

Table 2: Validation against grcwa / S4. The rightmost column reports the observed relative/absolute agreement; “exact” means agreement to the full printed precision of the reference value.

case	this work	reference	match
S4 golden T , p-pol	0.853173	0.852499*	$< 10^{-3}$
S4 golden T , s-pol	0.839654	0.839005*	$< 10^{-3}$
ex1 square holes (R, T)	0.132415, 0.867585	0.13241493, 0.86758507	✓
ex2 two layers, oblique (R, T)	0.132599, 0.867401	0.13259894, 0.86740106	✓
ex4 hexagonal (R, T)	0.164450, 0.835550	0.16445004, 0.83554996	✓
EUV mirror 13.5 nm R	0.599171	0.599171	exact
EUV mirror 13.7 nm R	0.647593	0.647593	exact
EUV absorber 13.5 nm R ($n = 97$)	0.578790	0.578790	exact
EUV absorber 13.5 nm R ($n = 9$)	0.566750	0.566750	exact
EUV absorber 13.5 nm, $\theta = 6^\circ$ ($n = 9$)	0.587072	0.587072	exact

* S4 golden value embedded in the regression test; grcwa itself computes 0.853173 / 0.839654, reproduced exactly by this work.

and the most recent S -matrix is memoized so that the reflectivity $\rightarrow$ field-reconstruction pattern does not rebuild $S(0, N-1)$.

- *Branch cut and field normalization.* The propagation constants are fixed to the physical sheet by $\Im \mathbf{q}_j < 0 \Rightarrow \mathbf{q}_j \mapsto -\mathbf{q}_j$, so evanescent modes decay in $+z$; the reconstructed transverse field keeps the $(E_x, E_y) \leftrightarrow (E_y, -E_x)$ rotation of (26), whose sign conventions are reproduced exactly in the $\mathbf{F}_{xy}$ block of Section 7.
- *Memory.* Uniform layers deduplicate their operators, and the block-wise post-processing of Section 7 keeps the peak footprint bounded by $O(n^2)$ rather than $O(n^2)$ at the larger $(4n)$ intermediate scale; measured peaks are reported in Table 8.

9 Results

9.1 Validation

Table 2 compares the solver with grcwa [9] and with values used by the S4 regression problem [3]. The grcwa examples are reproduced to the displayed precision; the C++ values are quoted to six digits here, but the underlying agreement is to the full double precision of the reference. Fourier coefficients of the reflected field and fields reconstructed in real space differ by approximately 10^{-13} in the tested cases. The separate S4 values are satisfied within the 10^{-3} relative tolerance prescribed by that regression test.

The square-lattice, two-layer, and hexagonal examples reproduce the grcwa reference values to 10^{-13} : the full-precision reflectivities are $R = 0.13241493181686462$ (ex1), $R = 0.1325989361682126$ (ex2, oblique incidence $\theta = \pi/10$), and $R = 0.16445003997328628$ (ex4, non-orthogonal hexagonal lattice). These exercise the circular (degeneracy-trimmed) harmonic selection, the anisotropic FFT convolution, oblique incidence, and the general non-Hermitian eigendecomposition, so the matching confirms not only the scattering assembly but also the reciprocal-lattice and convolution machinery.

The quasi-one-dimensional line grating (Section 9) is validated separately against grcwa at several truncations: $R = 0.528790, 0.528927, 0.529005$ for $n = 97, 199, 295$, in exact agreement with the reference at every order. In the reflected-field comparison the per-order Fourier coefficients (E_x, E_y, E_z) of the backward field agree with grcwa to a maximum absolute difference of 1.6×10^{-14} , the real-space $|E|^2$ grid to 3.4×10^{-14} , and the individual complex field components along the horizontal and vertical cell cross-sections to $2\text{--}4 \times 10^{-14}$ (normal incidence, $n = 9$). An oblique variant ($\theta = 6^\circ$) reproduces the specular and

all scattered orders per G -vector to machine precision. A broader sweep of the fast quasi-one-dimensional path over $\theta \in [0^\circ, 30^\circ]$, $\phi \in [0^\circ, 90^\circ]$, and both polarizations gives a worst-case relative field difference of $\sim 10^{-13}$ relative to *grcwa*, confirming that the per-harmonic reductions of Sections 3–5 hold at arbitrary incidence.

Finally, the integrated observables are validated: the absorbed-power integral $\text{Volume_integral}(\text{Re } \varepsilon^{-1})$ returns 0.196376 against *grcwa*'s 0.196392 ($\sim 8 \times 10^{-5}$ relative), and the three components of the z -stress-tensor integral equal the *grcwa* values ($-0.139217, -0.050671, -1.09948$) to the last digit. In addition to the fixed geometries, the lossless oblique stacks satisfy the energy-conservation invariant $R + T = \cos \theta$ to machine precision, which bounds the accumulation of error across the whole recursion.

9.1.1 Convergence with the harmonic count

For the EUV absorber on the Mo/Si mirror the reflectivity converges with the retained harmonic count n as

$$R = 0.5668, 0.5788, 0.5791, 0.57925 \quad \text{for } n = 9, 97, 197, 301, \quad (85)$$

respectively — approximately 0.06% above the $n \approx 100$ value at $n = 301$, consistent with the expected slow algebraic convergence of the Fourier series at the sharp absorber edges. The per-harmonic suffix (Section 3.2) keeps the solve cost linear in the number of uniform layers at every truncation, so the convergence scan is not dominated by the 80 Mo/Si interfaces. The quasi-one-dimensional grating converges similarly: $R = 0.528790, 0.528927, 0.529043, 0.529086, 0.529114$ at $n = 97, 199, 253, 359$ (normal incidence), with the conical counterpart ($\theta = 6^\circ, \phi = 45^\circ$) giving $R = 0.547546, 0.547773, 0.547820$ at $n = 113, 187, 253$.

9.2 Performance

The benchmark geometry uses wavelength $\lambda = 13.5$ nm and the layer sequence vacuum | TaN absorber | Ru cap | 40 periods of Mo/Si | Si substrate. Three geometric regimes are reported: (i) the *full-2D* EUV absorber, a 60×60 nm TaN rectangle on a 300 nm cell; (ii) the *quasi-1D* line grating, a 500 nm TaN bar spanning the cell at $L_x = 3.5$ μm ; and (iii) the conical (oblique) variants of both. Timings correspond to release-mode C++17 code linked to OpenBLAS and use six BLAS threads; each entry is the best measured wall time over repeated runs on the same workstation and is an implementation benchmark rather than a hardware-independent constant. The harmonic count n is the actual count after symmetry-preserving truncation, and all columns use the same n , material grids, incidence, and floating-point precision. In every mode the reflectivity R is bit-identical to the generic (unreduced) path, which is itself validated against *grcwa* in Section 9.

9.2.1 Effect of the per-harmonic uniform suffix (full 2D)

Table 3 isolates the per-harmonic uniform suffix (Section 3.2) on the full-2D EUV absorber, whose 40 Mo/Si periods contribute approximately 80 uniform interfaces. The suffix alone accelerates the S -matrix assembly by 3.4–3.7 $\times$ with bit-identical reflectivity; the gain grows with n because the suffix cost is now linear in n while the reference dense star products scale as $(2n)^3$. The residual dense work is a single patterned prefix (vacuum, absorber, cap) plus one cascade.

9.2.2 Quasi-one-dimensional line grating

For the y -invariant grating the combination of the one-dimensional harmonic selection (dropping the $y \neq 0$ dead modes), the per-harmonic suffix, the block-diagonal modal operator (Section 4.1), and the TE/TM-decoupled recursion (Section 5) applies at every interface. Table 4 reports end-to-end timings at normal

Table 3: Full-2D EUV absorber, S -matrix (`rt_solve`) before/after the per-harmonic uniform suffix (machine under load; R identical in both columns).

n	generic	per-harmonic suffix	speedup
201	1354 ms	369 ms	3.7×
301	3268 ms	964 ms	3.4×

Table 4: Quasi-1D EUV line grating, normal incidence ($\theta = \phi = 0^\circ$). Full TE/TM block-decoupled path: block-diagonal eigendecomposition plus block-decoupled S -matrix.

n	$2n$	setup (ms)	<code>rt_solve</code> (ms)	R
113	226	49	35	0.528927
187	374	122	102	0.529043
253	506	244	202	0.529086
359	718	579	452	0.529114

incidence; the setup column is dominated by the two $n \times n$ eigendecompositions and the `rt_solve` column by the $n \times n$ per-polarization recursion plus the cascade.

At conical incidence ($\theta = 6^\circ$, $\phi = 45^\circ$) the off-diagonal E_x – E_y coupling proportional to k_{y0} is physical, so the modal operator is no longer block diagonal. It is, however, block lower triangular (Section 4.2), so the eigenvalues still split into two n -dimensional problems and the eigenvectors are reconstructed by the Sylvester formula (70). Table 5 reports the resulting timings; the reconstruction is bit-identical to the full $2n \times 2n$ solve, so the reflectivity is unchanged and matches `grcwa` at every truncation.

9.2.3 Full two-dimensional geometry

For the genuinely two-dimensional absorber neither spectral reduction of the patterned operator applies, and the full $2n \times 2n$ non-Hermitian eigendecomposition remains the recognized bottleneck; only the per-harmonic uniform suffix (which is valid for any harmonic set) reduces the cost. Table 6 shows the resulting scaling. The total time is dominated by the patterned eigendecomposition at the largest n , confirming that the multilayer suffix — not the modal problem — was the historical wall for this class of stack.

9.2.4 Speedup summary and memory

Table 7 summarizes the isolated speedups of the individual reductions, and Table 8 the resulting memory footprint. Two memory reductions are relevant: uniform layers deduplicate their $(K, \mathbf{q}, \Phi)$ operators (one full matrix per distinct material plus a single shared identity Φ , rather than one copy per layer), and the block-wise post-processing of Section 7 avoids the $4n \times 4n$ and $3n \times 3n$ intermediate matrices. The persistent-storage reduction is approximately 20× on the periodic stacks; the measured peak resident set follows it closely, and both are validated against `/usr/bin/time` to within 5–9%.

9.2.5 Comparison with `grcwa`

Against the reference implementation, the full-2D EUV absorber ($n = 97$) runs the complete pipeline (FFT convolution, eigendecomposition, S -matrix, field reconstruction) in approximately 1.05 s versus `grcwa`’s 5.8 s — about 5× — with machine-precision agreement of every reflected order and the real-space field. The quasi-one-dimensional line grating ($n = 201$, normal incidence) is the most favorable case: the combined reductions give a 84 ms end-to-end solve, or approximately 80× relative to `grcwa`’s 11.0 s, at identical reflectivity $R = 0.528927$. These end-to-end factors include implementation-language and library effects

Table 5: Quasi-1D EUV line grating, conical incidence ($\theta = 6^\circ$, $\phi = 45^\circ$). Block-lower-triangular eigendecomposition (two $n \times n$ problems plus the coupled-eigenvector reconstruction (70)).

n	$2n$	setup (ms)	rt_solve (ms)	R
113	226	51	57	0.547546
187	374	123	204	0.547773
253	506	253	422	0.547820

Table 6: Full-2D EUV absorber, per-harmonic suffix path (general eigendecomposition retained).

mode	n	setup (ms)	rt_solve (ms)	R
normal	199	228	1309	0.528927
normal	497	2526	14402	0.529043
oblique ($\theta = 6^\circ$, $\phi = 45^\circ$)	199	257	1189	0.547546

(BLAS, FFTW, LAPACK) in addition to the algebraic reductions; the isolated reduction factors are those of Tables 3–7.

A further, independent gain is available from a threaded eigendecomposition. OpenBLAS executes `zgeev` single-threaded, whereas Intel MKL threads it internally: at $2n = 266$ MKL solves the patterned eigenproblem in ~ 80 ms against OpenBLAS’s ~ 126 ms ($1.6\times$), and the gap grows with size ($\sim 2\times$ at $2n = 530$). On the conical quasi-1D case ($n = 187$) the end-to-end time drops $\sim 1.5\times$ (1268 ms to ~ 830 ms) by switching the LAPACK backend alone. This is orthogonal to the reductions derived here: it accelerates the remaining dense eigendecomposition without changing its count.

10 Discussion

The per-harmonic uniform-suffix factorization and the exact eigenproblem decompositions are exact: they change only the *order* of arithmetic, not the model, and the scattering observables are bit-identical to the generic implementation. Three caveats bound their applicability.

First, the block-lower-triangular decomposition (Section 4.2) requires the harmonic set to be the one-dimensional row of a y -invariant grating, for which $\mathbf{k}_y$ is constant. For a genuinely two-dimensional pattern at oblique incidence the upper-right block is

$$\mathbf{M}_{12} = C \operatorname{diag}(\mathbf{k}_y) E \operatorname{diag}(\mathbf{k}_x) - \operatorname{diag}(\mathbf{k}_x \mathbf{k}_y), \quad (86)$$

which does not generally vanish. The implementation then uses the full $2n \times 2n$ problem; the choice is made from the norm of the computed block.

Second, the block-diagonal decomposition applies to the one-dimensional normal- incidence case and to additional patterns only when the computed off-diagonal blocks vanish. For a general two-dimensional pattern neither spectral reduction applies, and the full $2n \times 2n$ `zgeev` remains the recognized residual bottleneck. A matrix-free Krylov formulation (preconditioned by the quasi-1D solves above) is the natural next step for very large n , but is beyond the scope of this paper.

Third, the low-rank representation of Section 6 has not been used to replace the dense non-Hermitian eigensolver. Doing so requires a stability and conditioning analysis, particularly for lossy materials and near-degenerate modes. It should therefore be regarded as a direction for a reduced-order method rather than as part of the present exact algorithm.

Finally, we note that Li’s factorization rules [7] and the normal-vector method [8] improve *convergence* in n for high-contrast metallic gratings (reducing the required n), whereas the present reductions reduce the

Table 7: Isolated speedups of the individual exact reductions (mode-explicit; R bit-identical in every case).

case	before	after	factor
full-2D normal $n = 97$ <code>rt_solve</code>	963 ms	~300 ms	3.2×
full-2D normal $n = 201$ <code>rt_solve</code>	6420 ms	~1500 ms	4.2×
quasi-1D normal $n = 201$ total	~360 ms	84 ms	4.3×
quasi-1D normal $n = 253$ <code>rt_solve</code>	559 ms	202 ms	2.8×
quasi-1D oblique $n = 253$ setup (eig)	~605 ms	253 ms	2.4×

Table 8: Memory footprint (estimated peak from `--mem`, validated against `/usr/bin/time`; normal incidence).

config	persistent	<code>rt_solve</code> transient	peak RSS
quasi-1D $n = 113$	6.5 MiB	15.6 MiB	28 MiB
quasi-1D $n = 253$	32 MiB	78 MiB	142 MiB
full-2D $n = 199$	20 MiB	48 MiB	88 MiB

cost per harmonic. The two are orthogonal and compound: every harmonic saved by better factorization is processed at the reduced per-harmonic cost.

10.1 Future directions

Several directions follow directly from the operator structures exposed here.

Discrete point-group symmetry. The TE/TM separation of the y -invariant case is a continuous-symmetry consequence; a quasi-2D pattern instead retains discrete mirror symmetry $\varepsilon(x, -y) = \varepsilon(x, y)$. When the incidence is compatible ($k_{y0} = 0$) and the harmonic set is closed under the parity map $(i, j) \mapsto (i, -j)$, the convolution matrix C commutes with the corresponding permutation-parity operator, and a change of basis to even/odd combinations $|i, j\rangle_{\pm} = \frac{1}{\sqrt{2}}(|i, j\rangle \pm |i, -j\rangle)$ block-diagonalizes $\mathbf{M}$, $\mathbf{K}$, every interface matrix, and the S -matrix into two sectors of size $\approx n$. Each cubic operation then costs $2(n)^3 = \frac{1}{4}(2n)^3$; with the second mirror $\varepsilon(-x, y) = \varepsilon(x, y)$ the four parity sectors give $\frac{1}{16}(2n)^3$. This is an exact reduction in the same spirit as Sections 4.1–5, and its only cost is that the G -set must be a symmetry-closed (rectangular) product set.

Matrix-free single-excitation backend. The remaining asymptotic bottleneck is the dense $2n \times 2n$ non-Hermitian eigenproblem of a genuinely two-dimensional patterned layer, whose cost is cubic in n and whose storage is quadratic. For an aspect-ratio-anisotropic cell ($n_x \gg n_y$, e.g. a long line with y -features) a matrix-free boundary-value formulation is attractive: the uniform exterior and multilayer collapse to per-harmonic TE/TM port operators (Section 3), the patterned region is discretized in z by a spectral method, and the resulting linear system — of size $O(n_x n_y N_z)$, never assembled — is solved by a preconditioned Krylov method. The preconditioner is a block-diagonal collection of independent quasi-one-dimensional solves, one per y -channel with constant $k_y = k_{y0} + 2\pi m_y / L_y$, which the reductions of this paper already make cheap; the Krylov iteration then corrects only the genuinely two-dimensional coupling between y -orders. The convolution C is applied matrix-free through a pair of FFTs. This changes the scaling from $O((2n)^3)$ to $O(N_{\text{iter}} N_z n \log n)$ and is the natural successor to the dense modal backend at large n .

Adjoint gradients for inverse design. Because the S -matrix is excitation-independent, the physical (reciprocity) adjoint requires no differentiation of the eigendecomposition: the adjoint field is generated by an

adjoint incident amplitude vector against the already assembled S -matrix, and the per-pixel permittivity gradient is a real-space overlap of the forward and adjoint fields — the unconjugated z -integral analog of the block-wise post-processing in Section 7. This makes the eigenvalue phase/ordering ambiguity, which is a notorious difficulty of differentiable modal RCWA, irrelevant, and reuses the validated field reconstruction directly.

Reduced-rank eigensolver. The diagonal-plus-low-rank structure (77) of the binary-grating blocks has not been used to replace the dense eigensolver. A stable secular-equation or subspace-iteration method whose cost is $O(nr + r^3)$ rather than $O(n^3)$ — with a conditioning analysis for lossy and near-degenerate media — remains open and would compound with the reductions above for high-contrast step profiles.

11 Conclusion

We have identified invariant subspaces of the finite-dimensional Maxwell operator that remove avoidable dense algebra from Fourier-modal scattering. Homogeneous layers form a direct sum over diffraction harmonics, and a y -invariant patterned layer has an exactly reducible spectrum at both normal and conical incidence. These identities preserve the Fourier discretization and its physical observables. In the tested EUV stack, the homogeneous-suffix algorithm reduces scattering time by 3.4–3.7 for full two-dimensional truncations; additional one-dimensional reductions produce substantially larger gains for line gratings. The remaining asymptotic cost is the dense non-Hermitian modal problem of a genuinely two-dimensional patterned layer and the final coupling of that layer to its exterior channels. Matrix-free boundary formulations and structure-preserving iterative eigensolvers are the natural next steps.

References

- [1] M. G. Moharam and T. K. Gaylord. “Rigorous coupled-wave analysis of planar-grating diffraction”. In: *J. Opt. Soc. Am.* 71 (1981), pp. 811–818.
- [2] M. G. Moharam, E. B. Grann, D. A. Pommet, and T. K. Gaylord. “Formulation for stable and efficient implementation of the rigorous coupled-wave analysis of binary gratings”. In: *J. Opt. Soc. Am. A* 12 (1995), pp. 1068–1076.
- [3] V. Liu and S. Fan. “S4: a free electromagnetic solver for layered periodic structures”. In: *Comput. Phys. Commun.* 183 (2014), pp. 2233–2244.
- [4] D. M. Whittaker and I. S. Culshaw. “Scattering-matrix treatment of patterned multilayer photonic structures”. In: *Phys. Rev. B* 60 (1999), pp. 2610–2618.
- [5] L. Li. “Formulation and comparison of two recursive matrix algorithms for modeling layered diffraction gratings”. In: *J. Opt. Soc. Am. A* 13 (1996), pp. 1024–1035.
- [6] R. Redheffer. “On the relation of transmission-line theory to scattering and transfer”. In: *J. Math. Phys.* 41 (1962), pp. 1–41.
- [7] L. Li. “Use of Fourier series in the analysis of discontinuous periodic structures”. In: *J. Opt. Soc. Am. A* 13 (1996), pp. 1870–1876.
- [8] L. Li. “New formulation of the Fourier modal method for crossed surface-relief gratings”. In: *J. Opt. Soc. Am. A* 14 (1997), pp. 2758–2767.
- [9] *grcwa: rigorous coupled-wave analysis with reverse-mode automatic differentiation*. <https://github.com/tianliu-cs/grcwa>. Accessed: 2026.
- [10] D. Slepian. “Prolate spheroidal wave functions, Fourier analysis, and uncertainty—V: the discrete case”. In: *Bell Syst. Tech. J.* 57 (1978), pp. 1371–1430.